Calificación:

40% + 40% + 20%

1º parcial (3) — 2º parcial (3) — Entregas

Vamos a tener en cuenta la aceleración

Cargas periódicas, aunque hay algunas dinámicas que no son periódicas se podrán "hacer" periódicas

Se apoya en las ecuaciones del movimiento armónico simple (MAS)

$$EDO \rightarrow \frac{d^2x}{dt^2} + \omega_n^2 x = 0$$

Nomenclatura → Desplazamiento → x; Frecuencia natural → ω_n

$$x(t) = C_1 \cos(\omega_n t) + C_2 \operatorname{sen}(\omega_n t) \propto A \operatorname{sen}(\omega t + \varphi)$$

(1) Pto de la estructura donde $t=0 \rightarrow x = 0{,}37$ cm $\dot{x} = 0$ m/s. Frecuencia 0,25 H

a) $T = \frac{1}{f} = 4$ s b) $\omega = 2\pi f = 0{,}5\pi$ rad/s c) $A = x(t=0) = 0{,}37$ cm

d) $x(t) = A \operatorname{sen}(\omega t + \varphi) = A \cos(\omega t + \varphi)$

$x(0) = A \cos\varphi = 0{,}37 \cos\varphi \Rightarrow \varphi = 0° \Rightarrow x(t) = 0{,}37 \cos(0{,}5\pi t)$

e) $\dot{x}(t) = -0{,}58 \operatorname{sen}(\frac{\pi}{2} t)$ cm/s

g) $\dot{x}_{max} = 0{,}58$ cm/s g) $\ddot{x}(t) = -0{,}91 \cos(\frac{\pi}{2} t)$ cm/s² h) $\ddot{x}_{max} = 0{,}91$ cm/s²

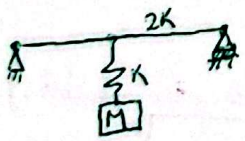
i) $x(t=3s) = 0{,}37 \cos(\frac{3\pi}{2}) = 0$ cm $\dot{x}(t=3s) = -0{,}58 \operatorname{sen}(\frac{3\pi}{2}) = 0{,}58$ cm/s

(3) Viga en voladizo con masa en el exterior ¿K_1 o K_2? $K_1 > K_2$

$E = K + U = \frac{1}{2}mv^2 + \cancel{\frac{1}{2}Kx^2}_{0} = \frac{1}{2}kA^2 \Rightarrow mv^2 = kA^2$ a mayor K mayor V; a igualdad de A

$\omega_n = \sqrt{\frac{K}{m}} \rightarrow T = \frac{2\pi}{\omega_n} > \frac{2\pi\sqrt{m}}{\sqrt{K}}$ A mayor K menor T

(4) $K = 10$ N/m $M = 200$ Kg



$\Rightarrow$ Están en serie

Escaneado con CamScanner

Usando la fórmula de Euler

$x(t) = Ae^{i(\omega t+\varphi)} = A(\cos(\omega_n t+\varphi) + i\,\mathrm{sen}(\omega_n t+\varphi))$

↳ $\underbrace{Ae^{i\varphi}}_{\bar{A}} \cdot e^{i\omega_n t} = \bar{A}e^{i\omega_n t}$

(14) $\omega = 10$ rad/s $\quad A = 5$ mm $\quad \varphi = \pi/6$

$\mathrm{Re}(\bar{A}) = A\cos\varphi = 5\cdot\cos\left(\frac{\pi}{6}\right) = \frac{5}{2}\sqrt{3}$

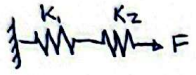
$\mathrm{Im}(\bar{A}) = A\,\mathrm{sen}\,\varphi = 5\,\mathrm{sen}\,\frac{\pi}{6} = \frac{5}{2}$

$\Rightarrow \bar{A} = \frac{5}{2}\sqrt{3} + i\frac{5}{2}$

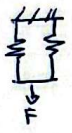
$x(t) = \bar{A}e^{i\omega_n t} = \left(\frac{5}{2}\sqrt{3} + i\frac{5}{2}\right)e^{i10t}$

$\dot{x}(t) = 25(-1+i\sqrt{3})\,e^{i10t} \qquad \ddot{x} = -100\left(\frac{5}{2}\sqrt{3} + i\frac{\sqrt{5}}{2}\right)e^{i10t}$

Muelles en serie y paralelo



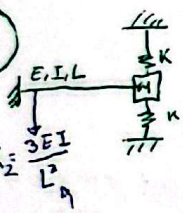
$K_{eq} = \frac{K_1 K_2}{K_1 + K_2}$



$K_{eq} = K_1 + K_2$

En función de como está cargada la estructura su rigidez cambia

(16)



$K_{eq} = K + K + K_2$

$\omega_n = \sqrt{\frac{2K + \frac{3EI}{L^3}}{M}}$

(17)



En serie $\quad K_{eq} = \frac{K_1 K_2}{K_1 + K_2}$

(19)

$F/\left(\frac{48EI}{L^3} + K\right) = 0{,}75x \Rightarrow F/\left(\frac{48EI}{L^3} + K\right) = 0{,}75\,\dot{x}$

$\left(\frac{48EI}{L^3} + K_i\right)\cdot\frac{1}{0{,}75} = \frac{48EI}{L^3} + K' \Rightarrow 1{,}1\cdot\frac{48EI}{L^3} + 1{,}1\,K_i = K' \Rightarrow$

$\Rightarrow K' = K_i + \Delta K = 0{,}33\left(\frac{48EI}{L^3} + 1{,}77\,K_i\right) \Rightarrow \frac{48EI}{L^3}0{,}33 + 0{,}33\,K_i$

Sistemas de un Grado de Libertad

- ¿Nº de grados de libertad? → GDL = (Nº masas) × (Cantidad de mov. posibles)

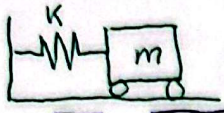
Aplicando la 2ª Ley de Newton $\boxed{m\frac{d^2x}{dt} + c\frac{dx}{dt} + Kx = F(t)}$

Sin amortiguamiento $m\ddot{x} + Kx = 0$ → $\begin{cases} \ddot{x} = \lambda^2 e^{\lambda t} \\ x = e^{\lambda t} \\ \dot{x} = \lambda e^{\lambda t} \end{cases}$ → $\lambda^2 = -\frac{K}{m}$ $\quad \lambda = \pm i\omega$, $\lambda^2 = -\omega^2$

Escaneado con CamScanner

$$\boxed{x = x_0 \cos \omega t + \frac{v_0}{\omega} \operatorname{sen} \omega t} \quad \text{con} \quad \omega = \sqrt{\frac{K}{m}}$$

* Particularizada en $t = 0$ *

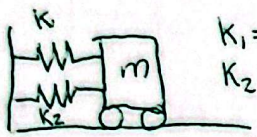
① 

$m = 20\ Kg$
$K = 400\ N/m$

$t = 0 \rightarrow \begin{cases} x_0 = 2mm \\ v_0 = 3\ m/s \end{cases}$

$$\omega_n = \sqrt{\frac{K}{m}} = \sqrt{\frac{400}{20}} = \sqrt{20} = 2\sqrt{5} = 4{,}5\ rad/s$$

$$x = 2\cos(4{,}5t) + \frac{3 \cdot 10^{-3}}{4{,}5} \operatorname{sen}(4{,}5t) \Rightarrow \underline{x(t) = 2\cos(4{,}5t) + 667{,}67 \operatorname{sen}(4{,}5t)}$$

② 

$K_1 = 1200\ N/m \quad m = 80\ Kg$
$K_2 = 400\ N/m$

$t = 5s \rightarrow \begin{cases} x = -2mm \\ v = -3m/s \end{cases}$

Muelles en paralelo $\Rightarrow K_{eq} = K_1 + K_2 = 1600\ N/m$

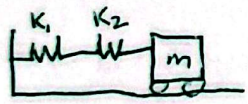
$$\omega_n = \sqrt{\frac{K_{eq}}{m}} = \sqrt{\frac{1600\ N/m}{80 Kg}} = 4{,}5\ rad/s$$

$x(t) = A\cos(\omega t) + B\operatorname{sen}(\omega t) \Rightarrow -2mm = A\cos(4{,}5 \cdot 5) + B\operatorname{sen}(4{,}5 \cdot 5)$ ①

$\dot{x}(t) = A\omega\operatorname{sen}(\omega t) + B\omega\cos(\omega t) \Rightarrow -3 \cdot 10^{-3} = A4{,}5\cos(4{,}5 \cdot 5) - B \cdot 4{,}5\operatorname{sen}(4{,}5 \cdot 5)$ ②

① → $-2 = -A \cdot 0{,}49 - B \cdot 0{,}87$
② → $-66{,}667 = -A\,0{,}87 + B\,0{,}49$

$$\Rightarrow \begin{cases} B = 59{,}16\ mm \\ A = -31{,}02mm \end{cases} \Rightarrow \underline{x(t) = 59{,}16\operatorname{sen}(4{,}5t) - 31{,}02\cos(4{,}5t)}$$

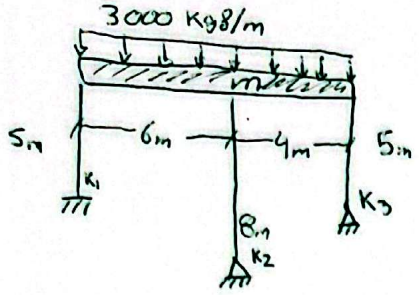
③ 

$K_{eq} = \frac{K_1 \cdot K_2}{K_1 + K_2} =$

$\omega_n = \sqrt{\frac{K}{m}}$

Escaneado con CamScanner

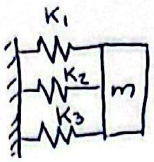
(7)



$E = 2 \cdot 10^6$ Kgf/cm²

$I = 3000$ cm⁴

$\propto$



$t = 2s$ → $x = 2$cm, ↳ $v = 4$m/s

$K_1 = \frac{12EI}{L^3} = 576$ Kgf/cm

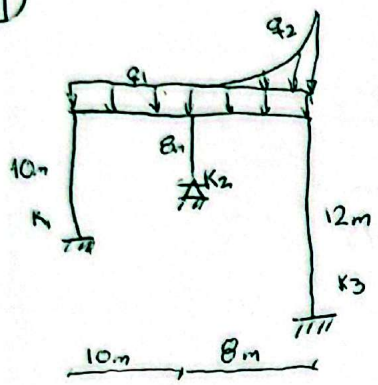
$K_2 = \frac{3EI}{L^3} = 35,16$ Kgf/cm

$K_3 = \frac{3EI}{L^3} = 144$ Kgf/cm

$K_{eq} = K_1 + K_2 + K_3 = 755,16$ Kgf/cm

$\omega_n = 4,97$ rad/s

(9)



$I_1 = I_3 = 4000$ cm⁴ $\quad I_2 = 3500$ cm⁴

$q_1 = 3500$ Kgf/m $\quad q_2 = 8e^x$

$E = 2 \cdot 10^6$ Kgf/cm² $\quad t = 5s$ → $x = 5$cm, ↳ $v = 2$m/s

$K_1 = \frac{12EI}{L^3} = 96$ Kgf/cm ✓

$K_2 = \frac{3EI}{L^3} = 41,02$ Kgf/cm ✓

$K_3 = \frac{12EI}{L^3} = 55,56$ Kgf/cm ✓

$\Rightarrow K_{eq} = 192,58$ Kgf/cm $= 1887,3$ N/cm $= 188\,728,4$ N/m

$F_{.1} = 3500 \cdot 18 = 63\,000$ Kgf $\Rightarrow 63\,000$ Kg $= m_1$

$F_2 = \int_0^8 8e^x dx = 8e^8 - 8 = 23839,66$ Kgf $\Rightarrow m_2 = 23839,66$ Kg

$m_{eq} = 86\,839,66$ Kg

$\omega_n = \sqrt{\frac{K_{eq}}{m_{eq}}} = 1,47$ rad/s ✓

$x(t) = A\cos(\omega_n t) + B\,\text{sen}(\omega_n t) \rightarrow 5 = A\cos(7,35) + B\,\text{sen}(7,35) = A \cdot 0,483 + B \cdot 0,876$ ①

$\dot{x}(t) = -A\omega_n \text{sen}(\omega_n t) + B\omega_n \cos(\omega_n t) \Rightarrow 200 = -A \cdot 1,29 + B \cdot 0,71$ ②

① → $A = \frac{5 - B \cdot 0,876}{5} = -116$ cm

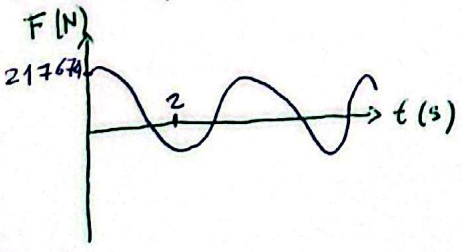
② → $200 = -1,29 + 0,226B - 0,71B \Rightarrow B = 70$ cm

$x(t) = -116\cos(1,47t) + 70\,\text{sen}(1,47t)$

$\dot{x}(t) = 170\,\text{sen}(1,47t) + 102,9\cos(1,47t)$

$\ddot{x}(t) = -A\omega_n^2 \cos(\omega_n t) - B\omega_n^2 \text{sen}(\omega_n t) \Rightarrow \dot{x}(t) = 250,66\cos(1,47t) - 151,26\,\text{sen}(1,47t)$ cm/s².

$F(t) = m_{eq}\ddot{x}(t) = 217674\cos(1,47t) - 131355\,\text{sen}(1,47t)$ N



Escaneado con CamScanner

+Péndulo

$$\theta = \theta_0 \cos\left(\sqrt{\frac{g}{L}}\,t\right) + \sqrt{\frac{L}{g}}\,\dot{\theta}_0 \operatorname{sen}\left(\sqrt{\frac{g}{L}}\,t\right)$$

Se puede usar la aproximación del péndulo físico

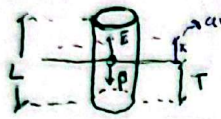
- Oscilación de cuerpo flotante

De normal Peso = Empuje. Si se le añade una fuerza Peso + Fuerza = Empuje, suelto y empiezan a oscilar

Con la dif de fuerzas se puede obtener una "K"

(11)

Ejercicio Spa:



Peso = Empuje (aumenta)

$$mg = (V_{sumerg}\,\rho_0)\,g \Rightarrow mg = (\pi r^2 T \rho_0)\,g$$

Desplazamos $x \rightarrow T' = T + x$

$$F_{neta} = \text{Peso} - E'_{\text{empuje}} \Rightarrow mg - (\pi r^2 (T+x)\rho_0)\,g = \underbrace{mg}_{\text{Peso}} - \underbrace{(\pi r^2 T \rho_0 g)}_{\text{Empuje}} - \pi r^2 x \rho_0 g \Rightarrow$$

$$\Rightarrow F_{neta} = \underbrace{-\pi r^2 \rho_0 g}_{K = \text{"muelle"}} \cdot x \rightarrow F = Kx$$

(m^2, kg/m^3, m/s^2)

$$\omega_n = \sqrt{\frac{K}{m}} = \sqrt{\frac{\pi r^2 \rho_0 g}{\pi r^2 L \rho}} \Rightarrow \boxed{\omega_n = \sqrt{\frac{\rho_0 g}{\rho L}} = \sqrt{\frac{\rho_0}{\rho}\cdot\frac{g}{L}}}$$

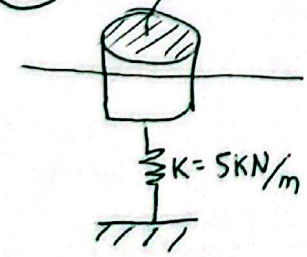
Frecuencia natural de algo flotando

ρ ↓ densid material

Esfera / Boya

$$V_{domen} = \frac{2}{3}\pi r^3$$

(12)



(m^2, kg/m^3, m/s^2)

$$K_{flotro} = A_T \rho_0 g = 0{,}7 \cdot 1000 \cdot 9{,}81 = 6867\ N/m$$

$$K_{muelle} = 5000\ N/m$$

Están en paralelo, el muelle se mueve a la vez que la bolla flotante, mismo $x \rightarrow K_{eq} = K_{flotro} + K_m = 11867\ N/m$

$$\omega_n = \sqrt{\frac{K}{m}} = \sqrt{\frac{11867\ N/m}{850\ Kg}} = 3{,}74\ rad/s$$

Escaneado con CamScanner

La frecuencia natural de un buque es:

$$\omega_n = \sqrt{\frac{hW}{J}}$$

W: weight
J: Inercia
h: distancia a G

+ Análisis con amortiguamiento

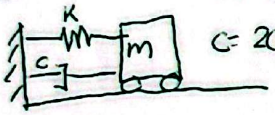
$$m\ddot{x} + c\dot{x} + Kx = 0 \Rightarrow \ddot{x} + \frac{c}{m}\dot{x} + \frac{K}{m} = 0 \Rightarrow \begin{cases} \omega_0^2 = K/m \\ c/m = 2\xi\omega_n \Rightarrow \xi = \frac{c}{2m\omega_n} \rightarrow \text{Razón de amortiguamiento} \end{cases}$$

- 3 tipos de amortiguamiento
  - Crítico $\xi = 1$
  - Superior al crítico $\xi > 1$
  - Inferior al crítico $\xi < 1$
    - → Trabajaremos aquí — Para $u(0) = u_0$ y $\dot{u}(0) = \dot{u}_0 \rightarrow$ $x(t) = e^{-\xi\omega_n t}\left(u_0 \cos\omega_a t + \frac{\xi\omega_n u_0 + \dot{u}_0}{\omega_a}\operatorname{sen}\omega_a t\right)$

$$\begin{cases} \xi < 1 \rightarrow x(t) = Be^{-\xi\omega_n t}\operatorname{sen}(\omega_a t + \varphi) \\ \xi = 1 \rightarrow x(t) = (B + Ct)e^{-\omega_n t} \\ \xi > 1 \rightarrow x(t) = Be^{(-\xi\omega_n + \omega_n\sqrt{\xi^2-1})t} + De^{(-\xi\omega_n - \omega_n\sqrt{\xi^2-1})t} \end{cases}$$

- $\omega_a = \omega_n\sqrt{1-\xi^2}$
- $\varphi = \arctan\left(\frac{x\cdot\omega_a}{\dot{x} + \xi\omega_n x}\right)$

(13)



$m = 800\,Kg$ $K = 100\,N/m$ $c = 20\,Ns/m$ $t = 4s$ $\begin{cases} x = 3mm \\ v = -5m/s \end{cases}$

$$\omega_n = \sqrt{\frac{K}{m}} = \sqrt{\frac{100}{800}} = 0,35\ rad/s$$

$$\xi = \frac{c}{2m\omega_n} = \frac{20}{2\cdot 800\cdot 0,35} = 0,036 < 1 \quad \text{Sist. subamortiguado}$$

$$x(t) = Be^{(-\xi\omega_n t)}\cdot\operatorname{sen}(\omega_a t + \varphi)$$

$$\omega_a = \omega_n\sqrt{1-\xi^2} = 0,35\sqrt{1-0,036^2} \approx 0,35$$

$$\varphi = \operatorname{arctg}\left(\frac{x\omega_a}{\dot{x} + \xi\omega_n x}\right) = \arctan\left(\frac{3\cdot 0,35}{5000 + 0,036\cdot 0,35\cdot 3}\right) = -2,1\cdot 10^{-4}\ rad$$

$t = 4$

$$3 = Be^{(-0,036\cdot 0,35\cdot 4)}\operatorname{sen}(0,35\cdot 4 - 2,1\cdot 10^{-4}) \Rightarrow B = 3,202\ mm$$

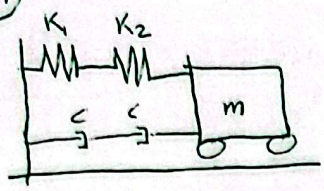
$$x(t) = 3,202\cdot e^{-0,0126t}\operatorname{sen}(0,35t - 2,1\cdot 10^{-4})\ [mm]$$

Escaneado con CamScanner

$x(t) = 3,62\cos(20,82t) - 8,91\,sen(20,82t) + 0,00086\cos(10t)$ [cm]

$\dot{x}(t) = -75,36\,sen(20,82t) - 185,5\cos(20,82t) - 0,086\,sen(10t)$ [cm/s]

(14)



$K_{eq} = \frac{K_1 \cdot K_2}{K_1 + K_2} = \frac{600 \cdot 300}{600+300} = 200$ N/m

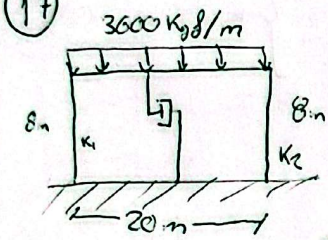
$C_{eq} = \frac{C_1 \cdot C_2}{C_1 + C_2} = \frac{80 \cdot 120}{80+120} = 48$ Ns/m

$\omega_n = \sqrt{\frac{K}{m}} = 0,63$ rad/s $\quad \zeta = \frac{c}{2m\omega_n} = 0,076 \quad \omega_a = \omega_n\sqrt{1-\zeta^2} = 0,61$

$t = 6s \quad x = 4mm \quad v = -2000$ mm/s $\quad \varphi = \arctan\left(\frac{4 \cdot 0,61}{-2000 + 0,076 \cdot 0,64 \cdot 4}\right) =$

$x(t) = Be^{-\zeta\omega_n t}\,sen(\omega_a t + \varphi)$

(17)



$m = 60.000$ Kg $= 61,1621$ Kgf s^2/cm

$I = 3500$ cm^4 $\quad E = 2 \cdot 10^6$ Kgf/cm^2 $\quad t = 2s \rightarrow x = -0,5$ cm, $v = 3$ m/s

$\zeta = 0,05$

$K_1 = K_2 = \frac{12EI}{L^3} = 169,0625$ Kgf/cm $\rightarrow K_{eq} = 2K_1$

$\omega = \sqrt{\frac{328,125}{61,1621}} = 2,5162$ rad/s

$\zeta = \frac{c}{2m\omega_n} \Rightarrow c = 14,1669$ Kgf s/cm $\Rightarrow \zeta < 1 \rightarrow$ Subamortiguado

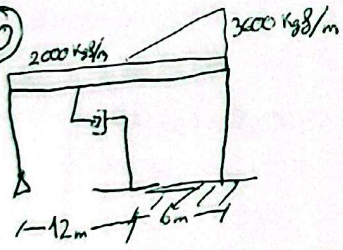
$x(t) = Be^{-\zeta\omega_n t}\,sen(\omega_a t + \varphi)$

$\omega_a = \omega_n\sqrt{1-\zeta^2} \quad \varphi = \arctan\left(\frac{x_0\omega_a}{\dot{x}_0 + \zeta\omega_n x_0}\right)$

$\vdots$

$x(t) = 0,6322\,e^{-0,1158t}\,sen(2,519t - 3,8889 \cdot 10^{-3})$

(20)



$K_1 = \frac{3EI}{L^3} = \frac{3 \cdot 2 \cdot 10^6 \text{ Kgf/cm}^2 \cdot 3500 \text{ cm}^4}{800^3 \text{ cm}^3} = 41,016$ Kgf/cm

$K_2 = \frac{12EI}{L^3} = \frac{12 \cdot 2 \cdot 10^6 \cdot 4000}{800^3} = 187,5$ Kgf/cm

$K_{eq} = 228,516$ Kgf/cm

$m = 2000 \text{ Kgf/m} \cdot 18 + \frac{3600 \cdot 6}{2} = 46800$ Kgf $= 47,706$ Kgf s^2/cm $\rightarrow \omega_n = 2,189$ rad/s

$\zeta = 0,06$

$x(t=5s) = -1,2$ cm $\quad \dot{x}(t=5s) = 3$ m/s $= 300$ cm/s

$\omega_a = \omega_n\sqrt{1-0,06^2} = 2,185$ rad/s $\quad \varphi = \arctan\left(\frac{-1,2 \cdot 2,185}{300 + 0,06 \cdot 2,189 \cdot (-1,2)}\right) = -8,74 \cdot 10^{-3}$

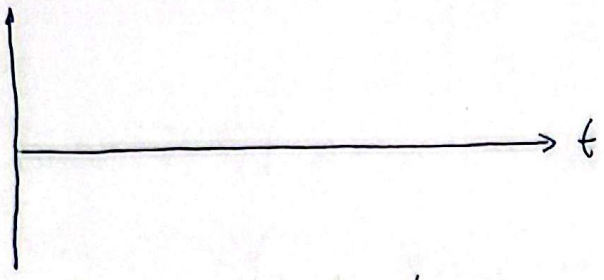
$B = \frac{-1,2}{e^{-0,06 \cdot 2,189 \cdot 5}\,sen(2,185 \cdot 5 - 8,74 \cdot 10^{-3})} = 2,32$ cm

$x(t) = 2,32\,e^{-0,13134t}\,sen(2,185t - 8,47 \cdot 10^{-3})$ [cm]

$\dot{x}(t) = -0,03\,e^{-0,13134t}\,sen(2,185t - 8,47 \cdot 10^{-3}) + 5,09e^{-0,13134t}\cos(2,185t - 8,47 \cdot 10^{-3})$ [cm/s]

Escaneado con CamScanner

$$\ddot{x}(t) = +0{,}039\, e^{-0{,}13134t}\, \text{sen}(2{,}185t - 8{,}74\cdot10^{-3}) - 0{,}0656\, e^{-0{,}13134t}\cos(2{,}185t - 8{,}74\cdot10^{-3})$$
$$-0{,}666\, e^{-0{,}13134t}\cos(2{,}185t - 8{,}74\cdot10^{-3}) - 11{,}076\, e^{-0{,}13134t}\,\text{sen}(2{,}185t - 8{,}74\cdot10^{-3})$$



+ Oscilaciones armónicas forzadas

En primera aproximación vamos a suponer una fuerza armónica → {Hace que la sig. ec. sea fácil} → $m\ddot{x} + Kx = F(t)$

$$x(t) = X_h + X_p = C_1\cos(\omega_n t) + C_2\,\text{sen}(\omega_n t) + \frac{F_0}{K - m\omega^2}\cos(\omega t)$$

Para $t=0$ y v_0, x_0

$$x(t) = x_0\cos(\omega_n t) + \frac{v_0}{\omega_n}\,\text{sen}(\omega_n t) + \frac{F_0}{K}\,\frac{1}{1-\left(\frac{\omega}{\omega_n}\right)^2}\left(\text{sen}(\omega t) - \left(\frac{\omega}{\omega_n}\right)\text{sen}(\omega_n t)\right)$$

(22) $m = 25$ Kg $\quad K = 6000$ N/m $\quad F(t) = 600\cos(20t)$

a) Cierta gráfica $t=0$; $x = 0{,}1$ m ; $\dot{x} = 2$ m/s)



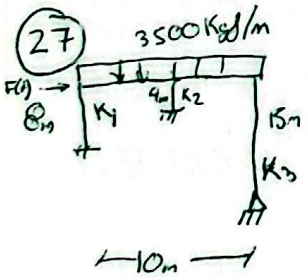
$$\omega_n = \sqrt{\frac{6000}{25}} = 15{,}49 \text{ rad/s}$$

$$x(t) = X_h(t) + X_p(t) = C_1\cos(15{,}49t) + C_2\,\text{sen}(15{,}49t) + \frac{600}{6000 - 25\cdot20^2}\cos(20t)$$

$$x(t=0s) = 0{,}1 = C_1 - 0{,}15$$

$$\dot{x}(t) = -15{,}49\,C_1\,\text{sen}(15{,}49t) - 15{,}49\,C_2\cos(15{,}49t) + 3\,\text{sen}(20t)$$

(27)



$$K_1 = \frac{2\cdot10^6\cdot80000\,12}{800^3} = 375 \text{ Kgf/cm}$$

$$K_2 = \frac{2\cdot10^6\cdot40000\cdot8}{400^3} = 15000 \text{ Kgf/cm}$$

$$K_3 = \frac{2\cdot10^6\cdot20000\cdot3}{1500^3} = 35{,}56 \text{ Kgf/cm}$$

$$K_{eq} = 15410{,}56 \text{ Kgf/cm}$$

1878

$$\omega_n = \sqrt{\frac{K}{m}} = 20{,}82 \text{ rad/s}$$

$$m = 35000 \text{ Kgf} = 35{,}678 \text{ Kgf s}^2/\text{cm}$$

$$x(t) = X_h(t) + X_p(t) = C_1\cos(20{,}82t) + C_2\,\text{sen}(20{,}82t) + \frac{1000\text{ N}}{(15410{,}56 - 35{,}67\cdot10^2)\cdot9{,}81 \text{ Kgf/cm}}\cos(10t) =$$

$$= C_1\cos(20{,}82t) + C_2\,\text{sen}(20{,}82t) + 0{,}0086\cos(10t)$$

$$\left.\begin{aligned} x_h(t=5) &= 0{,}4 = C_1\cdot(-0{,}91) + C_2(-0{,}415) \\ \dot{x}_h(t=5) &= 200 = C_1(+8{,}63) + C_2(-18{,}94) \end{aligned}\right\} \Rightarrow \begin{cases} C_1 = 3{,}62 \\ C_2 = -8{,}91 \end{cases}$$

Escaneado con CamScanner

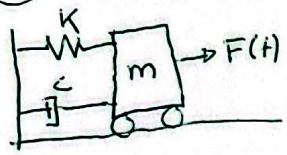
$$x(t) = 3{,}62\cos(20{,}82t) - 8{,}91\,\text{sen}(20{,}82t) + 0{,}0086\cos(10t)\ [cm]$$
$$\dot{x}(t) = -75{,}36\,\text{sen}(20{,}82t) - 185{,}5\cos(20{,}82t) - 0{,}086\,\text{sen}(10t)\ [cm/s]$$
$$\ddot{x}(t) = -1569\cos(20{,}82t) + 3862{,}11\,\text{sen}(20{,}82t) - 0{,}86\cos(10t)\ [cm/s^2]$$

• Con amortiguamiento

$$x_h(t)\begin{cases} -\zeta<1 \Rightarrow x_h(t) = Be^{-\zeta\omega_n t}\,\text{sen}\left(\omega_n\sqrt{1-\zeta^2}\,t + \arctan\left(\dfrac{x\cdot\omega_n\sqrt{1-\zeta^2}}{\dot{x}+\zeta\omega_n x}\right)\right) \\ -\zeta=1 \Rightarrow x_h(t) = (B+Ct)e^{-\omega_n t} \\ -\zeta>1 \Rightarrow x_h(t) = Be^{(-\zeta\omega_n+\omega_n\sqrt{\zeta^2-1})t} + De^{(-\zeta\omega_n-\omega_n\sqrt{\zeta^2-1})t} \end{cases}$$

$$x_p(t) \rightarrow \frac{}{\sqrt{(K-m\omega^2)^2 + C^2\omega^2}}\cos\left(\omega t - \arctan\left(\frac{C\omega}{K-m\omega^2}\right)\right)$$

(29)



$$\omega_n = \sqrt{\frac{K}{m}} = \sqrt{\frac{4000}{10}} = 20\ rad/s$$

$$\zeta = \frac{c}{2m\omega_n} = \frac{20}{2\cdot 10\cdot 20} = 0{,}05 < 1 \quad \text{subamortiguado}$$

$$x(t) = x_h(t) + x_p(t)$$

$$x_p = x\cos(\omega t + \varphi)$$

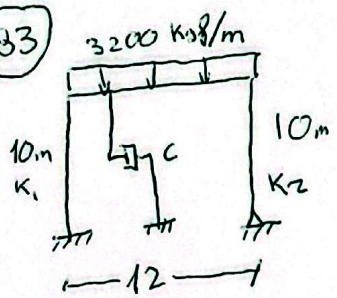
$$x = \frac{F_2}{\left[(K-m\omega^2)^2 + (C\omega^2)\right]^{1/2}} \qquad \varphi = \arctan\left(\frac{C\omega}{K-m\omega^2}\right)$$

$$x_h(t) = Be^{-\zeta\omega_n t}\,\text{sen}(\omega_n t + \varphi)$$

↓sub $\zeta<1$

Cond inis → $0{,}01 = Be^{0}(0 + \quad)$

(33)



$E = 2\cdot 10^6\ Kgf/cm^2$

$I_1 = 6000\,cm^4 \quad I_2 = 18000\,cm^4$

$F(t) = 5000\cos(30t)$

$\zeta = 0{,}2$

$$K_1 = \frac{2\cdot 10^6\ Kgf/cm^2 \cdot 6000\,cm^4 \cdot 12}{1000^3\,cm^3} = 144\ Kgf/cm$$

$$K_2 = \frac{3EI}{L^3} = 108\ Kgf/cm$$

$K_{eq} = 247212\ N/m$

$t = 4s \rightarrow x = 0{,}4\,cm$, $v = 2\,m/s$

$m = 3200\ Kgf/m \cdot 12m = 38400\ Kg$

$\zeta = \dfrac{c}{\quad}$

Escaneado con CamScanner

Ejercicio

5 olas: 2m; 3,5m; 1,2m; 4,1m; 0,8m ← En este caso

DFT $X[K] = \sum_{n=0}^{N-1} x[n]\, e^{-j\frac{2\pi K n}{N}}$

- $X[K]$: Secuencia de nombres complejos
- $x[n]$: Secuencia discreta de n puntos (Amplitudes registradas de las olas)
- Término oscilación → $K = 0,1,2,3,4$ Índice de frecuencia
- $N = 5$ (los puntos)

Para $K=0$

$$X[0] = \sum_{n=0}^{4} x[n]\, e^{-j\frac{2\pi 0 n}{5}} = \sum_{n=0}^{4} x[n]\cdot 1 = 2+3,5+1,2+4,1+0,8 = 11,16 + 0j$$

Para $K=1$

$$X[1] = \sum_{n=0}^{4} x[n]\, e^{-j\frac{2\pi n}{5}} = 2\cdot 1 + 3,5\, e^{-j\frac{2\pi}{5}\cdot 1} + 1,2\, e^{-j\frac{4\pi}{5}\cdot 1} + 4,1\, e^{-j\frac{6\pi}{5}\cdot 1} + 0,8\, e^{-j\frac{8\pi}{5}\cdot 1} = -0,159 - 0,863j$$

$\vdots$

$X[2] = 0,159 - 4,345j$

$X[3] = 0,159 + 4,35j$

$X[4] = -0,159 + 0,863j$

Calculamos en módulo de cada una:

$$\begin{cases} X[0] = 11,16 \\ X[1] = 1,29 \\ X[2] = 4,35 \\ X[3] = 4,35 \\ X[4] = 1,29 \end{cases}$$

Cada valor de $X[K]$ tiene una frecuencia asociada que llamaremos f_K

$f_K = \frac{K}{N\cdot \Delta t}$ → Índice de la DFT

↳ Nº total de ptos → Intervalo de tiempo entre cada medición [s]

$f_{K=0} = \frac{0}{5\cdot 1} = 0$ Hz $\quad f_{K=1} = \frac{1}{5} = 0,2$ Hz $\quad f_{K=2} = \frac{2}{5} = 0,4$ Hz $\quad f_{K=3} = \frac{3}{5} = 0,6$ Hz $\quad f_{K=4} = \frac{4}{5} = 0,8$ Hz

- Ecuación de la altura de ola

$$y(t) = A\cos(2\pi f t)$$

$y_1(t) = 11,16\cos(2\pi\cdot 0\cdot t) = 11,16$

$y_2(t) = 1,29\cos(2\pi 0,2t)$

$y_3(t) = 4,35\cos(2\pi 0,4t)$

$y_4(t) = 4,35\cos(2\pi 0,6\cdot t)$

$y_5(t) = 1,29\cos(2\pi 0,8\cdot t)$

Queremos pasar a fuerzas con la ecuación de Morison

$F(t) = F_{inercial}(t) + F_{arrastre}(t)$

- $F_{arrastre}(t)$ → Es proporcional al cuadrado de la velocidad, que depende de $\omega = 2\pi f$ ⇒ $A_{fuerza\ arrastre} \approx A_{ola}\cdot f$ (En 1ª aproximación despreciamos este)
- $F_{inercial}(t)$ → Depende {aceleración, frecuencia angular, amplitud de ola} → $\ddot{x} = A_{ola}\,\omega^2\cos(\omega t)$
  ↳ Dado que $F_{in} \propto \ddot{x}$ y amplitud de ola ⇒ $A_{fuerza\ inerc} \approx A_{ola}\cdot f^2$

Escaneado con CamScanner

$F_{din} \approx A_{fuerza} \cos(2\pi f t)$

$A_{fuerza} = C \cdot A_{ola} \cdot f^2$

→ Se estima en función de la geometría

En este caso $C = 1000$

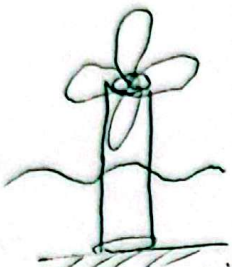
$f_1(t) = 0$

$f_2(t) = 1000 \cdot 1,29 \cdot 0,2^2 \cos(2\pi 0,2 t) = 0,0516 \cos(0,4\pi t)$ [N]

$f_3(t) = 1000 \cdot 4,35 \cdot 0,4^2 \cos(0,8\pi t) = 0,696 \cos(0,8\pi t)$ [N]

$f_4(t) = 1000 \cdot 4,35 \cdot 0,6^2 \cos(2\pi 0,6 t) = 1,566 \cos(1,2\pi t)$ [N]

$f_5 = 1000 \cdot 1,29 \cdot 0,8^2 \cos(2\pi 0,8 t) = 0,8256 \cos(1,6\pi t)$ [N]



Ppio superposición

$m\ddot{x} + c\dot{x} + Kx = f_1(t) + f_2(t) + f_3(t) + f_4(t) + f_5(t) \Rightarrow$

$\Rightarrow$ $m\ddot{x} + c\dot{x} + Kx = f_1(t) \Rightarrow x_1(t)$

$m\ddot{x} + c\dot{x} + Kx = f_2(t) \Rightarrow x_2(t)$

$\vdots \quad \Rightarrow \dots$

$m\ddot{x} + c\dot{x} + Kx = f_5(t) \Rightarrow x_5(t)$

$\rightarrow X(t) = x_1(t) + x_2(t) + x_3(t) + x_4(t) + x_5(t)$

Ejercicio Boya Tarifa

Alturas de ola: 0,84; 0,76; 0,75; 0,74; 0,76

10/10 - 00h - 04h

$\Delta t = 1h = 3600 s$

$X[0] = 0,84 + 0,76 + 0,75 + 0,74 + 0,76 = 3,85 + 0j = 3,85$

$X[1] = 0,1043 - 0,0059i = 0,1044$

$X[2] = 0,0707 + 0,0095i = 0,0714$

$X[3] = 0,0707 - 0,0951i = 0,0714$

$X[4] = 0,1043 + 0,0951i = 0,1044$

$f_k = \frac{K}{5 \cdot 3600} \Rightarrow$

$f_0 = 0$ Hz

$f_1 = 0,0556 \cdot 10^{-3}$ Hz

$f_2 = 0,1111 \cdot 10^{-3}$ Hz

$f_3 = 0,1667 \cdot 10^{-3}$ Hz

$f_4 = 0,222 \cdot 10^{-3}$ Hz

$\Rightarrow$

$y_1(t) = 3,85$ m

$y_2(t) = 0,1044 \cos(3,49 \cdot 10^{-4} t)$ m

$y_3(t) = 0,0714 \cos(6,98 \cdot 10^{-4} t)$ m

$y_4(t) = 0,0714 \cos(1,05 \cdot 10^{-3} t)$ m

$y_5(t) = 0,1044 \cos(1,4 \cdot 10^{-3} t)$ m

$F_{din_N} \approx C \cdot A_{ola} \cdot f_k^2 \cdot \cos(2\pi f_k t) \Rightarrow$

$f_1(t) = 0$ N

$f_2(t) = 0,323 \cdot 10^{-6} \cos(3,49 \cdot 10^{-4} t)$ N

$f_3(t) = 0,881 \cdot 10^{-6} \cos(6,98 \cdot 10^{-4} t)$ N

$f_4(t) = 1,98 \cdot 10^{-6} \cos(1,05 \cdot 10^{-3} t)$ N

$f_5(t) = 5,15 \cdot 10^{-6} \cos(1,4 \cdot 10^{-3} t)$ N

Escaneado con CamScanner

+ Cargas impulsivas

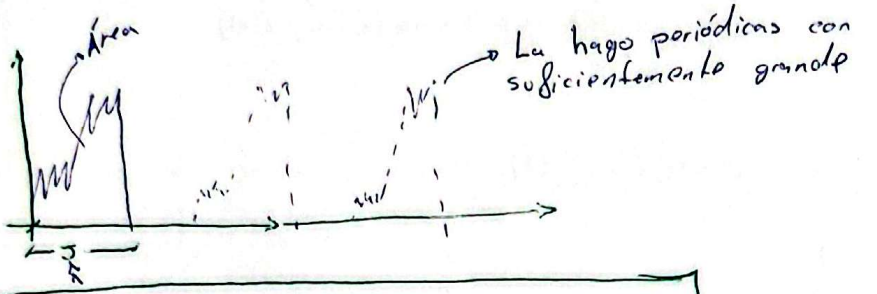
Fuerza muy grande (tiende ∞) y tiempo muy corto

El área debajo de esta fuerza es:

$$I = \lim_{\substack{f\to\infty \\ t\to 0}} \int_0^{\tau} f(t)\,dt$$

Hayamos la solución de la ecuación diferencial es:

$$x(t) = \frac{I}{\omega_n m} e^{-\zeta\omega_n t}\,\text{sen}(\omega_n t)$$



SISTEMAS DE DOS GRADOS DE LIBERTAD

GDL = N° masa × N° movim

Aparecen las matrices de masas y de rigidez $[M]$ y $[K]$, cuadradas de tamaño GDL × GDL

$[M][\ddot{x}] + [K][x] = 0$ → Por cada grado tenemos una frecuencia natural (autovalores / autovectores)

$[M] - \omega^2[X] + [K][x] = 0 \Rightarrow ([K] - \omega^2[M])[x] = [0]$

↳ Modos propios

Factor de participación modal

Represente el % de influencia de los modos

En términos porcentuales de la masa

Ejercicio 1

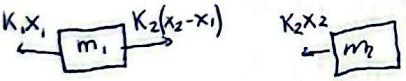


$K_1 = K_2 = K$

$m_1 = m_2 = m$

a) Frecs propias

$\det([K] - \lambda[M]) = 0$



$$\Rightarrow \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}\begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} + \begin{bmatrix} K_1+K_2 & -K_2 \\ -K_2 & K_2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\lambda = \omega^2 \qquad \begin{vmatrix} 2K - \lambda m & -K \\ -K & K - \lambda m \end{vmatrix} = 0 \Rightarrow m^2(\omega^2)^2 - 3Km\omega^2 + K^2 = 0 \Rightarrow (\omega^2)^2 - 3\frac{K}{m} + \left(\frac{K}{m}\right)^2 = 0$$

$$\Rightarrow \begin{cases} \lambda_1 = \omega_1^2 = \frac{3+\sqrt{5}}{2}\cdot\frac{K}{m} \\ \lambda_2 = \omega_2^2 = \frac{3-\sqrt{5}}{2}\cdot\frac{K}{m} \end{cases} \Rightarrow \begin{cases} \omega_{n1} = \sqrt{\frac{3+\sqrt{5}}{2}}\sqrt{\frac{K}{m}} \\ \omega_{n2} = \sqrt{\frac{3-\sqrt{5}}{2}}\sqrt{\frac{K}{m}} \end{cases}$$

Se ordenan de menor a mayor

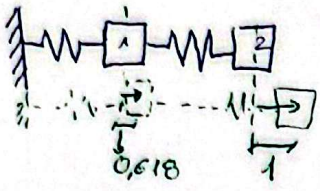
↳ MATLAB → eig(K; M)

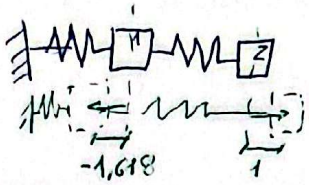
Escaneado con CamScanner

b) Autovectores / Modos propios

1º $\left(\begin{bmatrix} 2K & -K \\ -K & K \end{bmatrix} - \omega_1 \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \right) \begin{bmatrix} \phi_{11} \\ \phi_{12} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \dots \Rightarrow \phi_1 = \begin{bmatrix} \frac{\sqrt{5}}{2} - \frac{1}{2} \\ 1 \end{bmatrix} = \begin{bmatrix} 0{,}618 \\ 1 \end{bmatrix}$

2º $\left(\begin{bmatrix} 2K & -K \\ -K & K \end{bmatrix} - \omega_2 \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \right) \begin{bmatrix} \phi_{21} \\ \phi_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \dots \Rightarrow \phi_2 = \begin{bmatrix} -\frac{\sqrt{5}}{2} - \frac{1}{2} \\ 1 \end{bmatrix} = \begin{bmatrix} -1{,}618 \\ 1 \end{bmatrix}$





c) Factores de participación modal

$$m_{eff,i} = \frac{[\Phi]^T \cdot [M] \cdot [r]}{[\Phi]^T \cdot [M] \cdot [\Phi]}$$

$[\Phi] = [\phi_1 \; \phi_2]$

$[r] = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ Las 2 masas se mueven en paralelo

$[\Phi]^T \cdot [M][r] = [1{,}618 \; -0{,}618]$

$[\Phi]^T \cdot [M] \cdot [\Phi] = \begin{bmatrix} 1{,}3819 & 0 \\ 0 & 3{,}6179 \end{bmatrix}$

Suponemos $m = 1\,Kg \Rightarrow m_{TOT} = 2\,Kg$

$m_{eff,1} = \frac{1{,}618^2}{(+1{,}3819)} = 1{,}8944\,Kg \Rightarrow 94{,}72\%$

$m_{eff,2} = \frac{(-0{,}618)^2}{3{,}6179} = 0{,}1055\,Kg \Rightarrow 5{,}27\%$

Suma tiene que salir del 100%.

- Solución de la ecuación de libertad

$$x(t) = \sum_{j=1}^{N} \phi_j (a_j \cos(\omega_{nj} t) + b_j \cos(\omega_{nj} t))$$

$a = [\phi]^{-1} x_0 \quad ; \quad b = [\Omega_n]^{-1} [\phi]^{-1} \dot{x}_0$ siendo $[\Omega_n] = diag(\omega_{n1}, \omega_{n2} \dots \omega_{nN})$

- Ecuaciones de movimiento por solución directa:

siendo $x(0) = x_0 = \sum a_j \phi_j = \Phi a \Rightarrow a = [\Phi]^{-1} x_0$

Ejercicio 1:

masa 1: $x(0) = x_0 \; ; \; \dot{x}(0) = 0$

masa 2: $x(0) = 0 \; ; \; \dot{x}(0) = 0$

$a = \begin{bmatrix} \frac{\sqrt{5}}{2} & -\frac{\sqrt{5}}{2} - \frac{1}{2} \\ 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} x_0 \\ 0 \end{bmatrix} = \begin{bmatrix} \sqrt{5}/5 \\ -\sqrt{5}/5 \end{bmatrix} x_0 \; ; \; b = 0 \Rightarrow x(t) = X_1(t) + X_2(t)$

$\dot{x}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\Rightarrow$

Escaneado con CamScanner

$$\Rightarrow x(t)=\begin{bmatrix}x_1(t)\\x_2(t)\end{bmatrix}=\sum \phi_j\left(a\cos(\omega_{nj}t)+b\,\mathrm{sen}(\omega_{nj}t)\right)=\left(\underbrace{\begin{bmatrix}\frac{\sqrt{5}}{2}\\1\end{bmatrix}}_{\phi_1}\cdot\cos\left(\underbrace{\sqrt{\frac{3-\sqrt{5}}{2}}\sqrt{\frac{K}{m}}}_{\omega_{n1}}\cdot t\right)\right.-$$

$$\left.\underbrace{\begin{bmatrix}\frac{-\sqrt{5}}{2}\\1\end{bmatrix}}_{\phi_2}\cos\left(\underbrace{\sqrt{\frac{3+\sqrt{5}}{2}}\sqrt{\frac{K}{m}}}_{\omega_{n2}}t\right)\right)\cdot\frac{\sqrt{5}}{5}x_0$$

(−) constante a

factor común de a

+Solución por superposición modal

Permite desacoplar las ecuaciones de movimiento

Cogo un problema de "n" grados de libertad y lo trasforma en "n" problemas de 1 gdl

Cambiamos de coordenadas físicas a coordenadas modales

↳Cambio de variable $x=[\phi](\eta)$

multiplicamos por $[\phi]^T$

$$0=[M][\phi]\ddot{\eta}+[C][\phi]\dot{\eta}+[K][\phi]\eta=$$

$$[\phi]^T[M][\phi]\ddot{\eta}+[\phi]^T[C][\phi]\dot{\eta}+[\phi]^T[K][\phi]\eta=0\Rightarrow$$

Diagonal

$$\Rightarrow [M^*]\ddot{\eta}+[C^*]\dot{\eta}+[K^*]\eta=0$$

No tiene porque ser diagonal → No desacoplado

Amortiguamiento de Rayleigh $\Rightarrow [C^*]=\alpha[M^*]+\beta[K^*]$ → siempre va a ser diagonal

$\eta(t)$

Ya se pueden resolver los problemas de 1 gdl pero hay que deshacer el cambio de variable $\eta_j(t)=\eta_{j0}\cos(\omega_{nj}t)+\frac{\dot{\eta}_{j0}}{\omega_{nj}}\mathrm{sen}(\omega_{nj}t)$

superposición → $x(t)=\sum_{j=1}^{N}\left([\phi]\eta_j(t)\right)$

Ejercicio 1

Sabemos $[\phi]=\begin{bmatrix}\frac{\sqrt{5}-1}{2} & -\frac{\sqrt{5}+1}{2}\\1 & 1\end{bmatrix}$ $\quad \omega_n=\begin{cases}\sqrt{\frac{3-\sqrt{5}}{2}}\sqrt{\frac{K}{m}}\\ \sqrt{\frac{3+\sqrt{5}}{2}}\sqrt{\frac{K}{m}}\end{cases}$

$$[\phi]^T[M][\phi]=m\begin{bmatrix}\frac{5-\sqrt{5}}{2} & 0\\0 & \frac{5+\sqrt{5}}{2}\end{bmatrix}=[M^*]\qquad [\phi]^T[K][\phi]=\begin{bmatrix}5-2\sqrt{5} & 0\\0 & 2\sqrt{5}+5\end{bmatrix}=[K^*]$$

$x(0)=\begin{bmatrix}1\\0\end{bmatrix}$ % $\Rightarrow \dot{x}(t)=\begin{bmatrix}0\\0\end{bmatrix}$

$$\eta_0=[\phi]^{-1}x_0=\begin{bmatrix}\frac{\sqrt{5}-1}{2} & -\frac{\sqrt{5}+1}{2}\\1 & 1\end{bmatrix}\begin{bmatrix}x_0\\0\end{bmatrix}=\begin{bmatrix}\sqrt{5}/5\\-\sqrt{5}/5\end{bmatrix}x_0\qquad \dot{\eta}_0=[\phi]^{-1}\dot{x}_0^{\,0}=0$$

Escaneado con CamScanner

$$\eta(t) = \eta_{i0}\cos(\omega_{ni}t) + \frac{\dot{\eta}_{i0}}{\omega_{ni}}\operatorname{sen}(\omega_{ni}t)$$

$$\eta_1(t) = \frac{\sqrt{5}}{5}x_0\cos\left(\sqrt{\frac{3-\sqrt{5}}{2}}\sqrt{\frac{K}{m}}\cdot t\right) \quad ; \quad \eta_2(t) = -\frac{\sqrt{5}}{5}x_0\cos\left(\sqrt{\frac{3+\sqrt{5}}{2}}\sqrt{\frac{K}{m}}\,t\right)$$

Deshacemos el cambio de variable

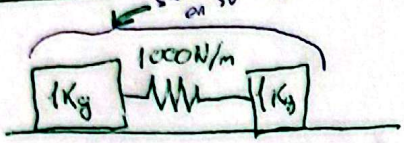
$$x(t) = [\phi]\cdot\eta(t) \Rightarrow x(t) = \begin{bmatrix}\frac{\sqrt{5}-1}{2} & -\frac{\sqrt{5}+1}{2}\\ 1 & 1\end{bmatrix}\begin{bmatrix}\eta_1(t)\\ \eta_2(t)\end{bmatrix}$$

Estructuras con modos rígidos

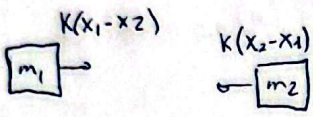
↳ Se mueve como sólido rígido

Ejercicio 2

Se mueve todo en su conjunto



$$[M] = \begin{bmatrix}m_1 & 0\\ 0 & m_2\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$



$$\Rightarrow [K] = \begin{bmatrix}K & -K\\ -K & K\end{bmatrix}$$

$$\begin{vmatrix}\lambda+K & -K\\ -K & \lambda+K\end{vmatrix} = 0 \Rightarrow \lambda^2 + 2K\lambda + K^2 - K^2 = 0 \Rightarrow \lambda^2 + 2K\lambda = 0 \Rightarrow$$

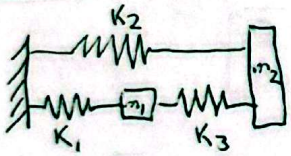
↳ $\lambda_1 = 0 \Rightarrow \omega_1 = 0$ rad/s (modo rígido)

↳ $\lambda_2 = -2K \Rightarrow \omega_2 = -10\sqrt{20}$

1º modo $\left(\begin{bmatrix}K & -K\\ -K & K\end{bmatrix} - \omega_1\begin{bmatrix}m & 0\\ 0 & m\end{bmatrix}\right)\begin{bmatrix}\phi_{11}\\ \phi_{12}\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix} \Rightarrow \begin{bmatrix}K & -K\\ -K & K\end{bmatrix}\begin{bmatrix}\phi_{11}\\ \phi_{12}\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix} \Rightarrow \phi_1 = 0$

2º modo $\left(\begin{bmatrix}K & -K\\ -K & K\end{bmatrix} - \omega_2\begin{bmatrix}m & 0\\ 0 & m\end{bmatrix}\right)\begin{bmatrix}\phi_{21}\\ \phi_{22}\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix} \Rightarrow \begin{bmatrix}K+10\sqrt{20}\,m & -K\\ -K & K+10\sqrt{20}\,m\end{bmatrix}\begin{bmatrix}\phi_{21}\\ \phi_{22}\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}$

Ejercicio 3



$m_1 = 2$ Kg $\quad m_2 = 1$ Kg $\quad K_1 = 1000$ N/m $\quad K_2 = 2000$ N/m $\quad K_3 = 3000$ N/m

$x_{10}(t=0) = x_0 \quad \dot{x}_1(t=0) = \dot{x}_2(t=0) = 0 \quad x_2(t=0) = 0$

Ec mov:

$$m_1\ddot{x}_1 = -K_1x_1 + K_3(x_2 - x_1)$$
$$m_2\ddot{x}_2 = -K_2x_2 - K_3(x_2 - x_1)$$
$$\Rightarrow \begin{bmatrix}m_1 & 0\\ 0 & m_2\end{bmatrix}\begin{bmatrix}\ddot{x}_1\\ \ddot{x}_2\end{bmatrix} + \begin{bmatrix}K_1+K_3 & -K_3\\ -K_3 & K_2+K_3\end{bmatrix} = 0$$

Frec's propias

$$\begin{vmatrix}K_1+K_3-\lambda m_1 & -K_3\\ -K_3 & K_2+K_3-\lambda m_2\end{vmatrix} = 0 \Rightarrow$$

↳ $\lambda_2 = 6098{,}1 \Rightarrow \omega_2 = 78{,}05$ rad/s

↳ $\lambda_1 = 901{,}9 \Rightarrow \omega_1 = 30{,}031$ rad/s

Modos ppios

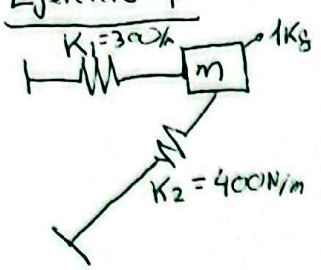
$$\Phi_1 = \begin{bmatrix}-0{,}366\\ 1\end{bmatrix} \quad \Phi_2 = \begin{bmatrix}1{,}36\\ 1\end{bmatrix}$$

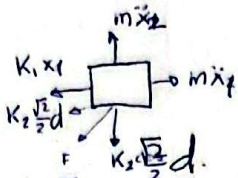
Escaneado con CamScanner

Factores de participación modal

$$m_{eff_i} = \frac{[\Phi]^T[M][\vec{0}]}{[\Phi]^T[M][\Phi]} = \frac{\begin{bmatrix} 0,268 \\ 3,72 \end{bmatrix}}{\begin{bmatrix} 1,628 & 0 \\ 0 & 4,7 \end{bmatrix}} \Rightarrow \begin{cases} meff_1 = 0,268^2/1,628 = 0,044 \\ meff_2 = 3,72^2/4,7 = 2,956 \end{cases}$$

Ejercicio 7





$$M = \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}$$

$$d = (x_1 \tfrac{\sqrt{2}}{2} + y \tfrac{\sqrt{2}}{2})$$

$$K = \begin{bmatrix} K_1 + \frac{K_2}{2} & \frac{K_2}{2} \\ K_2/2 & K_2/2 \end{bmatrix}$$

$$m\ddot{x}_2 = K_2 \tfrac{\sqrt{2}}{2} x_2$$

$$m\ddot{x}_1 = K_1 x_1 \tfrac{\sqrt{2}}{2} K_2 x_1$$

$$\left| \begin{pmatrix} 500 & 200 \\ 200 & 200 \end{pmatrix} + \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \right| = (500+\lambda)(200+\lambda) \cdot 200 \Rightarrow \begin{cases} \lambda_1 = 600 \\ \lambda_2 = 100 \end{cases}$$

$$\omega_1 = 10 \text{ rad/s} \qquad \omega_2 = 24,5 \text{ rad/s}$$

$$\left(\begin{bmatrix} 500 & 200 \\ 200 & 200 \end{bmatrix} - 100 \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \right) \begin{pmatrix} \phi_{11} \\ \phi_{12} \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} 566 & 0 \\ 0 & 266 \end{pmatrix} \begin{pmatrix} \phi_{11} \\ \phi_{12} \end{pmatrix} = 0$$

$$\Rightarrow \phi_1 = \begin{pmatrix} 1 \\ -2,13 \end{pmatrix}$$

$$\phi_2 = \begin{pmatrix} 1 \\ -2,16 \end{pmatrix}$$

$$m_{eff_i} = \frac{[\phi]^T \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} [\vec{1}]}{[\phi]^T \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} [\phi]} = \frac{\begin{bmatrix} 1 & -2,13 \\ 1 & -2,16 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}}{\begin{bmatrix} 1 & -2,13 \\ 1 & -2,16 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -2,13 & -2,16 \end{bmatrix}} = \frac{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}{\begin{bmatrix} 5,537 & 5,6 \\ 5,6 & 5,66 \end{bmatrix}} \Rightarrow$$

$$\Rightarrow meff_1 = 1^2/5,537 = 0,181$$

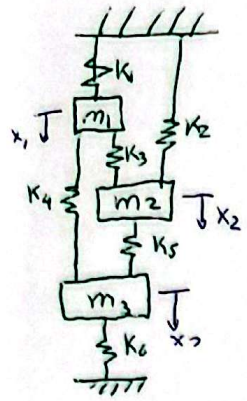
$$meff_2 = 0,177$$

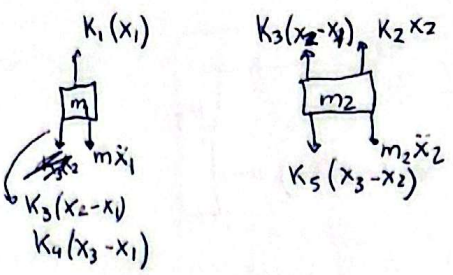
MAL La suma no da 1 Kg!!

Escaneado con CamScanner

Múltiples grados de libertad

Ejercicio 1





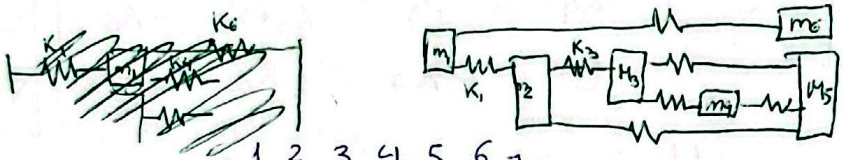
$$K=\begin{bmatrix} K_1+K_3+K_4 & -K_3 & -K_4 \\ -K_3 & K_2+K_3+K_5 & K_5 \\ -K_4 & K_5 & K_6+K_4+K_5 \end{bmatrix} \qquad m=\begin{bmatrix} m_1 & 0 & 0 \\ 0 & m_2 & 0 \\ 0 & 0 & m_3 \end{bmatrix}$$

$m_1 = 1\,Kg \quad m_2 = 2\,Kg \quad m_3 = 3\,Kg$

$K_N = N\ N/m$

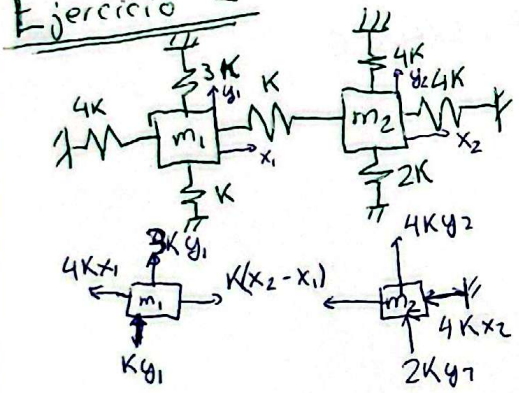
$$\Rightarrow \begin{cases} \omega_1 = 1{,}14\ rad/s \\ \omega_2 = 2{,}62\ rad/s \\ \omega_3 = 3{,}08\ rad/s \end{cases} \Rightarrow \begin{cases} \phi_1 = (-0{,}59, -0{,}6, -0{,}54) \\ \phi_2 = (0, -0{,}8, 0{,}6) \\ \phi_3 = (-0{,}46, 0{,}21, 0{,}14) \end{cases}$$

Ejercicio 4



$$[K_1] = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} K & -K & 0 & 0 & 0 & 0 \\ -K & K & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & & & & \\ & & & & & \\ & & & & & \end{bmatrix} \end{array} \qquad K$$

Ejercicio 5



$m_1 = m_2 = 1\,Kg$

$K = 1000\ N/m$

$$m\ddot{x}_1 = -4Kx_1 + K(x_2 - x_1) \qquad m\ddot{x}_2 = -K(x_2 - x_1) - 4Ky_2$$

$$m\ddot{y}_1 = -3Ky_1 - Ky_1 \qquad m\ddot{y}_2 = -4y_2 - 2Ky_1$$

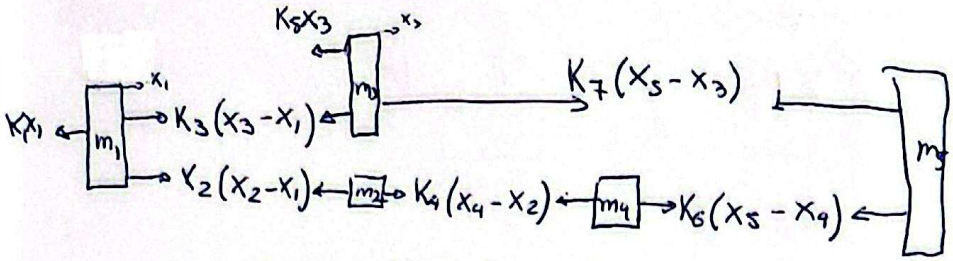
$$\begin{bmatrix} m_1 & 0 & 0 & 0 \\ & m_2 & 0 & 0 \\ & & m_1 & 0 \\ & & & m_2 \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{y}_1 \\ \ddot{y}_2 \end{bmatrix} + \begin{bmatrix} 5K & -K & 0 & 0 \\ -K & 5K & 0 & 0 \\ 0 & 0 & 4K & 0 \\ 0 & 0 & 0 & 6K \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

MATLAB → Frec's ppias y modos ppios

↳ [,] = eig(K, M)

Escaneado con CamScanner

Ejercicio 8



$$m_1\ddot{x}_1 = -K_1x_1 + K_3(x_3-x_1) + K_2(x_2-x_1)$$

$$m_2\ddot{x}_2 = -K_2(x_2-x_1) + K_4(x_4-x_2)$$

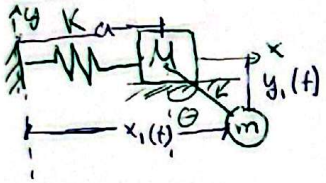
$$m_3\ddot{x}_3 = -K_5x_3 - K_3(x_3-x_1) + K_7(x_5-x_3)$$

$$m_4\ddot{x}_4 = -K_4(x_4-x_2) + K_6(x_5-x_4)$$

$$m_5\ddot{x}_5 = -K_7(x_5-x_3) - K_6(x_5-x_4)$$

$$\begin{bmatrix} m_1 & 0 & 0 & 0 & 0 \\ 0 & m_2 & 0 & 0 & 0 \\ 0 & 0 & m_3 & 0 & 0 \\ 0 & 0 & 0 & m_4 & 0 \\ 0 & 0 & 0 & 0 & m_5 \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \\ \ddot{x}_4 \\ \ddot{x}_5 \end{bmatrix} + \begin{bmatrix} K_1+K_3+K_2 & -K_2 & -K_3 & 0 & 0 \\ -K_2 & K_2+K_4 & 0 & -K_4 & 0 \\ -K_3 & 0 & K_5+K_3+K_7 & 0 & -K_7 \\ 0 & -K_4 & 0 & K_4+K_6 & -K_6 \\ 0 & 0 & -K_7 & -K_6 & K_7+K_6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = 0$$

Ejercicio 14



Lagrange $L = T - U$

Masa "m" del péndulo

$$\left.\begin{aligned} x_1 &= a + x + L\operatorname{sen}\theta \\ y_1 &= -L\cos\theta \end{aligned}\right\} \Rightarrow T_m = \frac{1}{2}m\left(\dot{x}_1^2 + \dot{y}_1^2\right)$$

$$T_m = \frac{1}{2}m\left((\dot{x} + L\dot{\theta}\cos\theta)^2 + (-L\dot{\theta}\operatorname{sen}\theta)^2\right) = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}mL^2\dot{\theta}^2 + mL\dot{x}\dot{\theta}\cos\theta$$

$$U_m = -mgy_1 = -mgL\cos\theta$$

Masa "M" de la torre

$$T_M = \frac{1}{2}M\dot{x}^2 \qquad U_M = \frac{1}{2}Kx^2$$

Total:

$$T_T = T_M + T_m = \frac{1}{2}(M+m)\dot{x}^2 + \frac{1}{2}mL^2\dot{\theta}^2 + mL\cos\theta\dot{x}\dot{\theta}$$

$$U_T = U_M + U_m = \frac{1}{2}Kx^2 - mgL\cos\theta$$

$$L = T_T - U_T = \frac{1}{2}(M+m)\dot{x}^2 + \frac{1}{2}mL^2\dot{\theta}^2 + mL\cos\theta\dot{x}\dot{\theta} - \frac{1}{2}Kx^2 + mgL\cos\theta$$

Para x: $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = 0 \Rightarrow (M+m)\ddot{x} + mL\cos\theta\ddot{\theta} - mL\operatorname{sen}\theta\dot{\theta}^2 + Kx = 0$

Para θ: $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0 \Rightarrow mL^2\ddot{\theta} + mL\cos\theta\ddot{x} + mgL\operatorname{sen}\theta$

Escaneado con CamScanner

Preguntas:

Tema 1:

- Desplaza x y se suelta => $x(t=0) = A = x$ → Sí

Tema 2:

- ¿Cuándo hay una fuerza F(t), primero se resuelve el sistema con las cond iniciales sin ~~nada~~ tener en cuenta la F y luego se le añade la x_p, que tiene en cuenta la F? → Sí

Tema 3

- Solución directa: ¿Tiene en cuenta el amortiguamiento? ¿Cómo? → Se desprecia
  - → Da varias ec's de desplaz. son las de cada masa?? ¿Dirección? ¿Se suman??
- Superposición modal: Resolver n problemas de 1GDL. La wn es de esos nuevos problemas?? Y modos?? Sí
  - La vector
  - Puede tener en cuenta C

Tema 4:

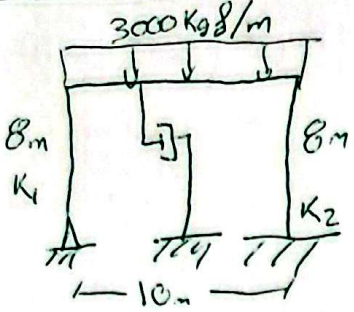
- ¿Qué pasa si uno de los Fact. de participación modal < 0? No debería

Tema 5

- Ejercicio 8: ¿Matriz de masas? → Como lo hice, ver problemas

Escaneado con CamScanner

Tema 2: Ejercicio 19



$E = 2 \cdot 10^6 \text{ Kgf/cm}^2$ $\quad t = 2s \begin{cases} x = 0,5 cm \\ v = 3 m/s \end{cases}$ $\quad \xi = 0,05$?? No ostn enunc.

$I = 2000 cm^4$

$$K_1 = \frac{3EI}{L^3} = \frac{3 \cdot 2 \cdot 10^6 \text{ Kgf/cm}^2 \cdot 2000 cm^4}{800^3 cm^3} = 23,44 \text{ Kgf/cm}$$

$$K_2 = \frac{12EI}{L^3} = 93,75 \text{ Kgf/cm}$$

$$K_T = K_1 + K_2 = 117,19 \frac{\text{Kgf}}{cm} \cdot \frac{100 cm}{1 m} \cdot \frac{9,8 N}{1 \text{ Kgf}} = 114963,39 \text{ N/m}$$

$m = 3000 \text{ Kgf/m} \cdot 10 m = 30.000 \text{ Kgf} = 30.000 \text{ Kg}$ $\quad \omega_n = \sqrt{\frac{K}{m}} = 1,96 \text{ rad/s}$

$\omega_n = \omega_n \sqrt{1 - \xi^2} = 1,955 \text{ rad/s}$

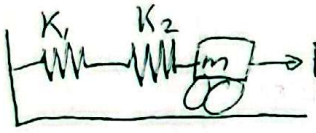
$$\varphi = \arctan\left(\frac{0,005 \cdot 1,955}{3 + 0,05 \cdot 1,96 \cdot 0,005}\right) = 0,00326 \text{ rad}$$

$\xi < 1$ Subamortig $\Rightarrow x(t) = Be^{-\xi \omega_n t} \operatorname{sen}(\omega_n t + \varphi) \Rightarrow B = \dfrac{0,005}{e^{-0,05 \cdot 1,96 \cdot 2} \operatorname{sen}(1,955 \cdot 2 + 0,00326)} \Rightarrow$

$\Rightarrow B = -0,0087 m = -0,87 cm$

$x(t) = -0,87 e^{-0,098t} \operatorname{sen}(1,955t + 3,26 \cdot 10^{-3}) \text{ cm}$

Tema 2 Ejercicio 23



$m = 35 \text{ Kg}$ $\quad K_1 = 5000 \text{ N/m}$ $\quad K_2 = 7000 \text{ N/m}$

$F(t) = 250 \cos(25t)$ $\quad t = 2s \begin{cases} x = 0,03 mm \\ v = 1,5 m/s \end{cases}$

$$K_{eq} = \frac{K_1 K_2}{K_1 + K_2} = 2916,67 \text{ N/m} \Rightarrow \omega_n = \sqrt{\frac{K}{m}} = 9,13 \text{ rad/s}$$

$$\left.\begin{aligned} x(t) &= C_1 \cos(\omega_n t) + C_2 \operatorname{sen}(\omega_n t) - \\ \dot{x}(t) &= -C_1 \omega_n \operatorname{sen}(\omega_n t) + C_2 \omega_n \cos(\omega_n t) \end{aligned}\right\} \Rightarrow \left\{\begin{aligned} 0,03 \cdot 10^{-3} &= 0,831 C_1 - 0,556 C_2 \\ 1,5 &= -5,08 C_1 + 7,59 C_2 \end{aligned}\right\} \Rightarrow$$

$$\Rightarrow \begin{cases} C_1 = 0,24 \\ C_2 = 0,36 \end{cases}$$

$$x(t) = 0,24 \cos(9,13t) + 0,36 \operatorname{sen}(9,13t) + \frac{250}{2916,67 - 785 \cdot 25^2} \cos(25t) \Rightarrow$$

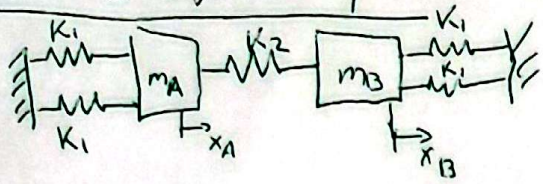
$$\Rightarrow x(t) = 0,24 \cos(9,13t) + 0,36 \operatorname{sen}(9,13t) - 0,013 \cos(25t)$$

Escaneado con CamScanner

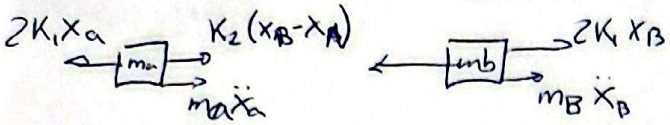
Tema 3: Ejercicio 4

$K_1 = 1000\ N/m$ $K_2 = 500\ N/m$ $m_a = m_b = 2\ Kg$

$t=0$ $x_a = x_0$ x_b $v_a = v_b = 0$



$$[M] = \begin{pmatrix} m_a & 0 \\ 0 & m_b \end{pmatrix} \quad [K] = \begin{pmatrix} 2K_1+K_2 & -K_2 \\ -K_2 & 2K_1+K_2 \end{pmatrix}$$



$$\det([K] - \lambda[M]) = 0 \Rightarrow \begin{vmatrix} 2500-2\lambda & -500 \\ -500 & 2500-2\lambda \end{vmatrix} = 0$$

$$\Rightarrow (2500-2\lambda)^2 - 500^2 = 0 \Rightarrow \begin{cases} \lambda_1 = 1000 \\ \lambda_2 = 1500 \end{cases} \Rightarrow \begin{cases} \omega_1 = 31,6\ rad/s = \\ \omega_2 = 38,73\ rad/s = \end{cases}$$

1er modo ω_1

$$([K] - \omega_1[M])(\phi) = 0 \Rightarrow \begin{bmatrix} \cdot & -1 \\ \cdot & \cdot \end{bmatrix} \phi_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

2º modo

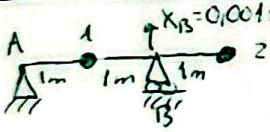
$$\begin{bmatrix} 2500-2\cdot 38,73 & -500 \\ -500 & 2500-2\cdot 38,73 \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix} = 0 \Rightarrow \phi_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\Rightarrow \begin{cases} 2422,54\phi_1 - 500\phi_2 = 0 \\ -500\phi_1 + 2422,54\phi_2 = 0 \end{cases} \Rightarrow \phi_1 = 1 \Rightarrow \phi_2 = 2422,54/500 =$$

$$[\phi]^T[M][r] = (4, 0) \rightarrow m_{eeg1} = \frac{18}{4} = 4\ Kg = 100\%$$

$$[\phi]^T[M][\phi] = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} \quad m_{eeg2} = \frac{0}{4} = 0 = 0\%$$

Tema 3: Ejercicio 10



$m_1 = m_2 = 0,1\ t\ s^2/m$ $EI = 3000\ t/m^2$ $\Rightarrow [K] = \begin{bmatrix} 28800 & 7200 \\ 7200 & 4800 \end{bmatrix}$

$E = 3\cdot 10^6\ ton/m^2$ $\rightarrow [M] = \begin{bmatrix} 9,8 & 0 \\ 0 & 9,8 \end{bmatrix}$

$I = 0,0001\ m^4$

$$\Phi = \begin{bmatrix} -0,277 & 3,6103 \\ 1 & 1 \end{bmatrix} \quad \omega_n \begin{cases} 16,9203 \\ 56,056 \end{cases} \quad [M^*] = [\Phi][M][\Phi] = \begin{bmatrix} 10,552 & 0 \\ 0 & 137,54 \end{bmatrix}$$

$$[K^*] = \begin{bmatrix} 3020 & 0 \\ 0 & 432160 \end{bmatrix}$$

$$\eta_0 = [\phi]^{-1}[x_0] = [\phi]^{-1}\begin{bmatrix} 0 \\ 0,0001\,sen(200\cdot 0) \end{bmatrix} = 0$$

$$\dot{\eta}_0 = [\phi]^{-1}[\dot{x}_0] = [\phi]^{-1}\begin{bmatrix} 0 \\ 0,2 \end{bmatrix} = \begin{bmatrix} 0,1857 \\ 0,0143 \end{bmatrix} \Rightarrow \eta(t) = \frac{\dot{\eta}_{j0}}{\omega_j}\cos(\omega_j t) + \frac{\dot{\eta}_j}{\omega_j}sen(\omega_j t)$$

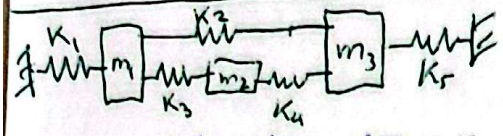
$$\eta_1(t) = \frac{0,1857}{16,92}\,sen(16,92\,t) \qquad \eta_2(t) = \frac{0,0143}{56,056}\,sen(56,056\,t)$$

$$x(t) = \begin{bmatrix} -0,277 & 3,6103 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 0,0109\,sen(16,92t) \\ 2,55\cdot 10^{-4}\,sen(56,056t) \end{bmatrix}$$

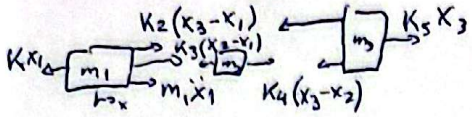
Escaneado con CamScanner

Tema 1: Ejercicio 5

Tema 4: Ejercicio 3



$$[M]=\begin{bmatrix} m_1 & 0 & 0 \\ 0 & m_2 & 0 \\ 0 & 0 & m_3 \end{bmatrix}$$



$$[K]=\begin{bmatrix} K_1+K_2+K_3 & -K_3 & -K_2 \\ -K_3 & K_3+K_4 & -K_4 \\ -K_2 & -K_4 & K_2+K_4+K_5 \end{bmatrix}$$

$\omega_1 = 29,47$ rad/s $\quad \phi_1 = \begin{pmatrix} 1,25 \\ 1,47 \\ 1 \end{pmatrix}$

$\omega_2 = 71,221$ rad/s $\quad \phi_2 = \begin{pmatrix} -0,98 \\ -0,81 \\ 1 \end{pmatrix}$

$\omega_3 = 85$ rad/s $\quad \phi_3 = \begin{pmatrix} -21,84 \\ 8,28 \\ 1 \end{pmatrix}$

$m_T = m_1+m_2+m_3 = 6$ Kg

$$[\phi]^T[M]\begin{bmatrix}1\\1\\1\end{bmatrix} = \begin{pmatrix} 7,1972 \\ 6,8912 \\ -2,3385 \end{pmatrix}$$

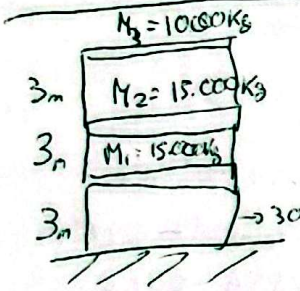
$$[\phi]^T[M][\phi] = \begin{pmatrix} 8,91 & 0 & 0 \\ 0 & 4,56 & 0 \\ 0 & 0 & 619,56 \end{pmatrix}$$

$m_{eeq1} = 7,2^2/8,91 = 6,31$ Kg $\sim 100\%$

$m_{eeq2} = 0,89^2/4,56 = 0,17$ Kg $\sim 0\%$

$m_{eeq} = 2,342/619,56 = 0,0088$ Kg $\sim 0\%$

Tema 4 Ejercicio 10



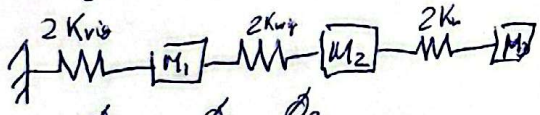
$E = 2,1 \cdot 10^7$ KN/m2

$\Rightarrow I = \frac{0,3 \cdot 0,3^3}{12}$

$K_{viga} = \frac{12EI}{3^3} = \frac{12 \cdot 2,1 \cdot 10^7 \cdot \frac{0,3 \cdot 0,3^3}{12}}{27} \Rightarrow$

$\Rightarrow K_{viga} = 6300$ N/m

$$[M] = \begin{bmatrix} 15000 & 0 & 0 \\ 0 & 15000 & 0 \\ 0 & 0 & 10000 \end{bmatrix}$$



$$[K] = \begin{bmatrix} 25200 & -12600 & 0 \\ -12600 & 25200 & -12600 \\ 0 & -12600 & 12600 \end{bmatrix}$$

$\phi_1 \quad \phi_2 \quad \phi_3$

$$\Phi = \begin{bmatrix} 0,4772 & -1,034 & 0,896 \\ 0,8396 & -0,2031 & -1,36 \\ 1 & 1 & 1 \end{bmatrix}$$

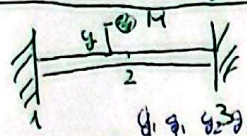
$$\omega_n = \begin{cases} \omega_1 = 0,4496 \\ \omega_2 = 1,2312 \\ \omega_3 = 1,7035 \end{cases} \Rightarrow \begin{cases} m_{eff1} = 36898 = 92,25\% \\ m_{eff2} = 2783,7 = 6,96\% \\ m_{eff3} = 318,3 = 0,8\% \end{cases}$$

$$\eta_0 = [\phi]^{-1} X_0 = \begin{pmatrix} 0,2984 & 0,525 & 0,468 \\ -0,5812 & -0,114 & 0,373 \\ 0,2828 & -0,4114 & 0,212 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

↳ Cond. iniciales abst

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Tema 5: Ejercicio 8



K_{12} =

	y_1	g_1	y_2	g_2
y_1	$\frac{12EI}{L^3}$	$\frac{6EI}{L^2}$	$-\frac{12EI}{L^3}$	$\frac{6EI}{L^2}$
g_1	$\frac{6EI}{L^2}$	$\frac{4EI}{L}$	$-\frac{6EI}{L^2}$	$\frac{2EI}{L}$
y_2	$-\frac{12EI}{L^3}$	$-\frac{6EI}{L^2}$	$\frac{12EI}{L^3}$	$-\frac{6EI}{L^2}$
g_2	$\frac{6EI}{L^2}$	$\frac{2EI}{L}$	$-\frac{6EI}{L^2}$	$\frac{4EI}{L}$

K_{13} es equivalente

$[M]$ =

	y_1	g_1	y_2	g_2	y_3	g_3
y_1						
g_1						
y_2			M			
g_2						
y_3						
g_3						

K_{TOT} =

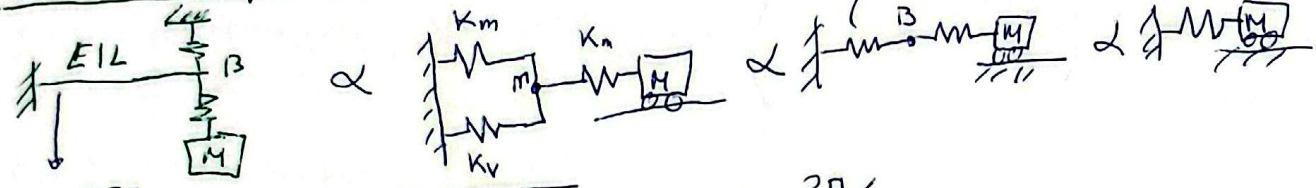
	y_1	g_1	y_2	g_2	y_3	g_3
y_1		$3L$	-6	$3L$	0	0
g_1	$3L$	$2L^2$	$-3L$	L^2	0	0
y_2	-6	$-3L$	12	0	-12	6
g_2	$6L$	$2L^2$	0	$8L^2$	$-6L$	$2L^2$
y_3	-6					
g_3						

emp 1

emp 2.

$\Rightarrow K_{corr} = \begin{bmatrix} 12 & 0 \\ 0 & 8L^2 \end{bmatrix} \frac{EI}{L^3}$

Tema 1: Ejercicio 18

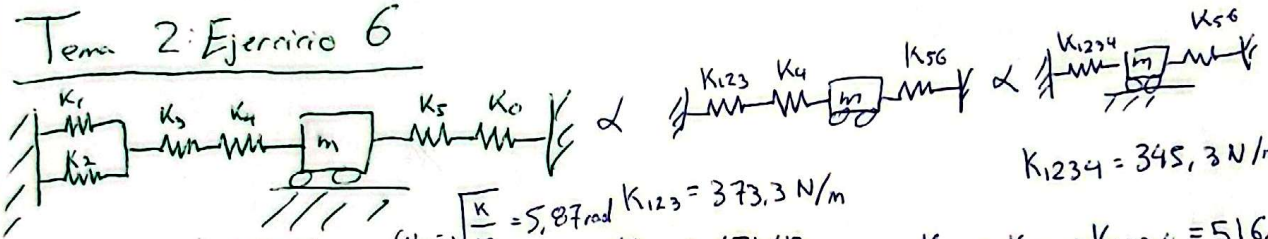


$K_v = \frac{3EI}{L^3}$

$\omega = \sqrt{\frac{(K_m+K_v)K_m}{(2K_m+K_v)M}}$

$T = 2\pi/\omega$

Tema 2: Ejercicio 6



$m = 15\ kg$ $\quad K_4 = 4600\ N/m$

$K_1 = 300\ N/m$ $\quad K_5 = 200\ N/m$

$K_2 = 400\ N/m$ $\quad K_6 = 1200\ N/m$

$K_3 = 800\ N/m$ $\quad t = 3s \begin{cases} x = 0,015m \\ \dot{x} = -6m/s \end{cases}$

$\omega_n = \sqrt{\frac{K}{m}} = 5,87\ rad$ $\quad K_{123} = 373,3\ N/m$ $\quad K_{56} = 171,43$

$K_{1234} = 345,3\ N/m$

$K_{eq} = K_{56} + K_{1234} = 516,7\ N/m$

$x(t) = A\cos(5,87t) + B\sin(5,87t)$

$\dot{x}(t) = -A5,87\sin(5,87t) + B5,87\cos(5,87t)$

$\Rightarrow \begin{cases} 0,015 = 0,325A + 0,996B \\ -6 = 5,55A + 1,91B \end{cases}$

$\Rightarrow \begin{cases} A = -0,962 \\ B = -0,1346 \end{cases}$

Tema 1: Ejercicio 2

$t=0 \rightarrow x=-A$

$K = 15791,361\ N/m$ $\quad m = 4\ kg$ $\quad x = 0,04\ m \rightarrow \dot{x} = 5,76\ m/s$

a) ¿A?

$E_T = \frac{1}{2}KA^2 = E_c + E_p = \frac{1}{2}mv^2 + \frac{1}{2}Kx^2 \Rightarrow \frac{15791,361}{2}A^2 = \frac{4}{2}(5,76)^2 + \frac{15791,361}{2}(0,04)^2 \Rightarrow A = 0,1\ m$

b) $¿E_m? = \frac{1}{2}KA^2 = 78,96\ J$

c) $¿V_{max}? \rightarrow E_m = E_c + E_p^{\,0} \Rightarrow 78,96\ J = \frac{1}{2}4 \cdot v_{max}^2 \Rightarrow V_{max} = 6,28\ m/s$

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Tema 1: Ejercicio 5

$m = 500\ kg$ $A = 2m$ $\omega = 2\ rad/s$

$t = 0 \rightarrow x = 0$ pos eq

a) E_c's pos y vel

$t=0, x=0$

$x(t) = A \operatorname{sen}(\omega_n t + \varphi) \Rightarrow 0 = A \operatorname{sen}\varphi \Rightarrow \operatorname{sen}\varphi = 0 \Rightarrow \varphi = 0\ rad$

$x(t) = 2 \operatorname{sen}(2t)\ m$; $\dot{x}(t) = 4\cos(2t)\ m/s$

b) $E_p(t)$ y $E_c(t)$ $\omega = \sqrt{K/m} \Rightarrow K = 2000\ N/m$

$E_p = \frac{1}{2} K (x(t))^2 = 1000\,(2\operatorname{sen}(2t))^2 = 4000\operatorname{sen}^2(2t)\ J$

$E_c = \frac{1}{2} m (\dot{x}(t))^2 = 250\,(4\cos(2t))^2 = 4000\cos^2(2t)\ J$

c) $t \rightarrow E_{p\,max}$ y $E_{c\,max}$

$E_{p_{max}}$ $\operatorname{sen}(2t) = 1 \Rightarrow 2t = \frac{\pi}{2} + 2\pi K$ $(K = 0, 1 \dots) \Rightarrow t = \frac{\pi}{4} + \pi K$ $(K = 0, 1, 2 \dots)\ s$

$E_{c\,max}$ $\cos(2t) = 1 \Rightarrow 2t = 0 + 2\pi K$ $(K = 0, 1, \dots) \Rightarrow t = \pi K$ $(K = 0, 1, \dots)\ s$

Tema 1 Ejercicio 7

$m = 30\ Kg$ $A = 50\ cm$ $f = 4\ Hz \Rightarrow \omega = 2\pi f = 8\pi\ rad/s$

$x(t) = A \operatorname{sen}(\omega_n t + \varphi) \Rightarrow t = 0\ s$ $x = A \Rightarrow A = A \operatorname{sen}(8\pi \cdot 0 + \varphi) \Rightarrow 1 = \operatorname{sen}(\varphi) \Rightarrow \varphi = \frac{\pi}{2}$

$x(t) = 0{,}5 \operatorname{sen}(8\pi t + \frac{\pi}{2})\ m$; $\dot{x}(t) = 4\pi\cos(8\pi t + \frac{\pi}{2})\ m/s$; $\ddot{x}(t) = -32\pi^2 \operatorname{sen}(8\pi t + \frac{\pi}{2})\ m/s^2$

$\ddot{X}_{max} = 32\pi^2 = 315{,}83\ m/s^2 \Rightarrow F_{max} = m \cdot a_{max} = 9474{,}8\ N$

$T = 2\pi/\omega = 1/f = 0{,}25\ s$; $\omega = \sqrt{K/m} \Rightarrow K = 18949{,}6\ N/m$

$E_T = \frac{1}{2} K A^2 = 2368{,}7\ J$; $E_K = \frac{1}{2} m \cdot V_{max}^2 = \frac{1}{2} \cdot 30 \cdot (4\pi)^2 = 2368{,}7\ J$ ($x = 0$, $V = V_{max}$); $E_p = E_T - E_K = 0\ J$

Tema 1 Ejercicio 8

$M = 200\ g$ (0,2 Kg) $K = 20\ N/m$ $A = 5\ cm$ (0,05 m) $\omega = \sqrt{\frac{K}{m}} = 10\ rad/s$

$x(t) = 0{,}02 = 0{,}05 \operatorname{sen}(10t + \varphi) \Rightarrow \operatorname{sen}(10t + \varphi) = 0{,}4 \Rightarrow 10t + \varphi = 0{,}4115$

$\dot{x}(t) = 0{,}05 \cdot 10 \cos(10t + \varphi) \Rightarrow \dot{x}(t) = 0{,}5\cos(0{,}4115) = 0{,}458\ m/s$

$E_K + E_p = E_T \Rightarrow 3E_p = E_T \Rightarrow 3 \cdot \frac{1}{2} K x = \frac{1}{2} K A^2 \Rightarrow x = 0{,}00833\ m = 0{,}083\ cm$ (↳ $E_K = 2E_p$)

Tema 1 Ejercicio 10

$m = 2\ Kg$ $K = 4{,}5\ N/m$ $x = 0 \Rightarrow E_K = 0{,}9 \cdot 10^{-3}\ J$ $\omega = \sqrt{K/m} = 1{,}5\ rad/s$

$x(t=0) = 0 = A \operatorname{sen}(1{,}5 \cdot 0 + \varphi) \Rightarrow \operatorname{sen}(\varphi) = 0 \Rightarrow \varphi = 0$

$E_K + E_p = E_T \Rightarrow 0{,}9 \cdot 10^{-3} = \frac{1}{2} K A^2 \Rightarrow A = 0{,}02\ m$ (pto eq $\Rightarrow x = 0 \Rightarrow E_p = 0$)

$\Rightarrow x(t) = 0{,}02 \operatorname{sen}(1{,}5t)\ m$; ↳ $\dot{x}(t) = 0{,}03\cos(1{,}5t)$

$E_K = \frac{1}{2} m (\dot{x}(t))^2 = 3 \cdot 10^{-4} \Rightarrow 0{,}0009\cos^2(1{,}5t) = 0{,}0003 \Rightarrow \cos(1{,}5t) = \frac{1}{\sqrt{3}} \Rightarrow$

$\Rightarrow 1{,}5t = 0{,}9553 + 2\pi K$ $(K = 0, 1, 2 \dots) \Rightarrow t = 0{,}637 + 1{,}33\pi K$ $(K = 0, 1, 2 \dots)$

Escaneado con CamScanner

Tema 1: Ejercicio 12

$m = 50g$ $K = 2000 N/m$ $A = 2cm = 0{,}02m$

" $0{,}5Kg$

$\omega = \sqrt{\frac{K}{m}} = 20\ rad/s \Rightarrow T = 2\pi/\omega = \pi/10\ s = 0{,}31\ s$

$\hookrightarrow f = \omega/2\pi = \frac{10}{\pi} Hz = 3{,}18\ Hz$

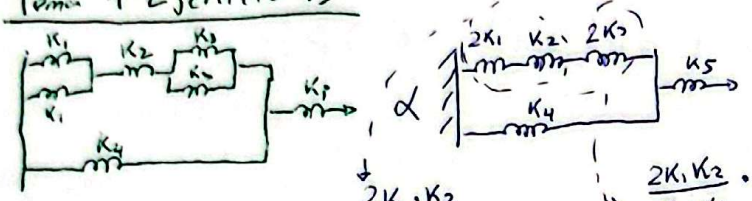
$x(t) = A\,sen(\omega t + \varphi) \Rightarrow sen(\varphi) = 1 \Rightarrow \varphi = \frac{\pi}{2} \Rightarrow x(t) = 0{,}02\,sen\left(20t + \frac{\pi}{2}\right) m$

(t=0 x=A)

$\dot{x}(t) = 0{,}4\cos\left(20t + \frac{\pi}{2}\right) \Rightarrow \dot{x}_{max} = 0{,}4\ m/s$

$\ddot{x}(t) = -8\,sen\left(20t + \frac{\pi}{2}\right) \Rightarrow \ddot{x}_{max} = 8\ m/s^2$

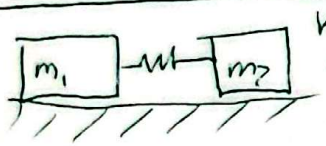
Tema 1: Ejercicio 15



$\frac{2K_1 \cdot K_2}{K_1 + K_2}$ $\qquad$ $\frac{\frac{2K_1K_2}{K_1+K_2} \cdot 2K_3}{\frac{2K_1K_2}{K_1+K_2} + 2K_3} + K_4$

$$K_{eq} = \frac{\left[\left(\frac{\frac{2K_1K_2}{K_1+K_2} \cdot 2K_3}{\frac{2K_1K_2}{K_1+K_2} + 2K_3}\right) + K_4\right] \cdot K_5}{\left[\left(\frac{\frac{2K_1K_2}{K_1+K_2} \cdot 2K_3}{\frac{2K_1K_2}{K_1+K_2} + 2K_3}\right) + K_4\right] + K_5}$$

Tema 3: Ejercicio 2



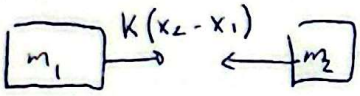
$K = 1000 N/m$

$m_1 = m_2 = 1Kg$

$x_1(t=0) = x_0 \qquad \dot{x}_2(t=0) = \dot{x}_1(t=0) = \dot{x}_2(t=0) = 0$

$M = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$[K] = \begin{bmatrix} K & -K \\ -K & K \end{bmatrix} = \begin{bmatrix} 1000 & -1000 \\ -1000 & 1000 \end{bmatrix}$



$|K - \omega^2 M| = 0 \Rightarrow \omega^2 = \lambda = \begin{cases} \omega_1 = 0\ rad/s \\ \omega_2 = 44{,}72\ rad/s \end{cases} \Rightarrow \phi_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \phi_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$meeB_1 = 2^2/2 = 2Kg\ 100\% \qquad meeB_2 = 0^2/2 = 0Kg = 0\%$

$$x(t) = \sum_{j=1}^{N} \phi_j \left(a_j \cos(\omega_{nj} t) + \cancel{b_j \cos(\omega_{nj} t)}^{0}\right)$$

$a = [\phi]^{-1} X_0 = \begin{bmatrix} 0{,}5 & 0{,}5 \\ -0{,}5 & 0{,}5 \end{bmatrix} \begin{bmatrix} x_0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0{,}5 \\ -0{,}5 \end{bmatrix} x_0$

$b = [\Omega]^{-1}[\phi]^{-1} \dot{x}_0 = \begin{bmatrix} 0 & 0 \\ 0 & 44{,}72 \end{bmatrix}^{-1} \begin{bmatrix} 0{,}5 & 0{,}5 \\ 0{,}5 & 0{,}5 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0$

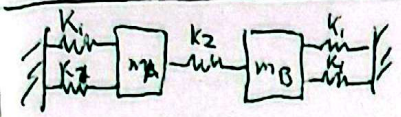
$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \underbrace{\begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{\phi_1} \underbrace{\cos(\cancel{0t})}_{1} \cdot \underbrace{0{,}5 x_0}_{a_1} + \underbrace{\begin{pmatrix} -1 \\ 1 \end{pmatrix}}_{\phi_2} \cos(44{,}72t) \cdot \underbrace{(-0{,}5 x_0)}_{a_2} = \begin{pmatrix} 0{,}5 x_0 \\ 0{,}5 x_0 \end{pmatrix} + \begin{pmatrix} x_0 \cos(44{,}72t) \\ -x_0 \cos(44{,}72t) \end{pmatrix}$$

$\omega_1 \qquad \omega_2$

$$\dot{x}(t) = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{pmatrix} -x_0\, 44{,}72\, sen(44{,}72t) \\ x_0\, 44{,}72\, sen(44{,}72t) \end{pmatrix} \qquad \ddot{x}(t) = \begin{pmatrix} -x_0\, 44{,}72^2 \cos(44{,}72t) \\ x_0\, 44{,}72^2 \cos(44{,}72t) \end{pmatrix}$$

Escaneado con CamScanner

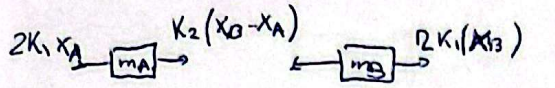
Tema 3: Ejercicio 4



$x_A(t=0) = x_0$; el resto $= 0$

$m_A = m_B = 2\,Kg$

$K_1 = 1000 \quad K_2 = 500 \qquad [M] = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$



$$[K] = \begin{bmatrix} 2K_1 + K_2 & -K_2 \\ -K_2 & 2K_1 + K_2 \end{bmatrix} = \begin{bmatrix} 2500 & -500 \\ -500 & 2500 \end{bmatrix}$$

$$|[K] - \lambda[M]| = 0 \Rightarrow \begin{vmatrix} 2500 - 2\lambda & -500 \\ -500 & 2500 - 2\lambda \end{vmatrix} = 0 \Rightarrow \lambda = \omega^2 \Rightarrow \begin{cases} \omega_1 = 31,62 \text{ rad/s} \\ \omega_2 = 38,73 \text{ rad/s} \end{cases}$$

$$([K] - \omega^2[M]) \begin{pmatrix} \phi_{x1} \\ \phi_{x2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \phi_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} ; \phi_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$m_{eff_i} = \frac{[\phi]^T [M] \begin{pmatrix} 1 \\ 1 \end{pmatrix}}{[\phi]^T [M] [\phi]} = \frac{\begin{pmatrix} 4 \\ 0 \end{pmatrix}}{\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}} \Rightarrow \begin{cases} m_{eff_1} = 4^2/4 = 4\,Kg = 100\% \\ m_{eff_2} = 0^2/4 = 0\,Kg = 0\% \end{cases}$$

Método directo

$$a = [\phi]^{-1} x_0 = \begin{pmatrix} 0,5 & 0,5 \\ -0,5 & 0,5 \end{pmatrix} \begin{pmatrix} x_0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0,5 \\ -0,5 \end{pmatrix} x_0$$

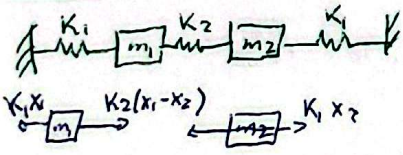
$$b = [\Omega]^{-1} [\phi]^{-1} \dot{x}_0 = \begin{bmatrix} 31,62 & 0 \\ 0 & 38,73 \end{bmatrix}^{-1} \begin{pmatrix} 0,5 & 0,5 \\ -0,5 & 0,5 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0$$

$$x(t) = \sum_{j=1}^{N} \phi_j \left(a_j \cos(\omega_{nj} t) + b_j \operatorname{sen}(\omega_{nj} t) \right) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} (0,5 x_0 \cos(31,62t) + 0) + \begin{pmatrix} 1 \\ -1 \end{pmatrix} (-0,5 x_0 \cos(38,73t) + 0)$$

$$\Rightarrow x(t) = \begin{pmatrix} x_A(t) \\ x_B(t) \end{pmatrix} = \begin{pmatrix} 0,5 x_0 \cos(31,62t) - 0,5 x_0 \cos(38,73t) \\ 0,5 x_0 \cos(31,62t) + 0,5 x_0 \cos(38,73t) \end{pmatrix}$$

$$\dot{x}(t) = \frac{\partial x(t)}{\partial t} \quad ; \quad \ddot{x}(t) = \frac{\partial \dot{x}(t)}{\partial t}$$

Tema 3: Ejercicio 5



$m_1 = m_2 = 1\,Kg \qquad [M] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$K_1 = 400\,N/m$

$K_2 = 250\,N/m \qquad [K] = \begin{bmatrix} K_1 + K_2 & -K_2 \\ -K_2 & K_1 + K_2 \end{bmatrix} = \begin{bmatrix} 650 & -250 \\ -250 & 650 \end{bmatrix}$

$\omega_1 = 20 \text{ rad/s} \qquad \phi_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\omega_2 = 30 \text{ rad/s} \qquad \phi_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$$m_{eff} = \frac{[\phi]^T [M] \begin{pmatrix} 1 \\ 1 \end{pmatrix}}{[\phi]^T [M] [\phi]} = \frac{\begin{pmatrix} 2 \\ 0 \end{pmatrix}}{\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}}$$

$m_{eff_1} = 2^2/2 = 2\,Kg = 100\%$

$m_{eff_2} = 0^2/2 = 0\,Kg = 0\%$

$$a = [\phi]^{-1} \begin{pmatrix} x_0 \\ 0 \end{pmatrix} = \begin{bmatrix} 0,5 & 0,5 \\ 0,5 & 0,5 \end{bmatrix} \begin{pmatrix} x_0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0,5 \\ -0,5 \end{pmatrix} x_0$$

$$b = [\Omega]^{-1} [\phi]^{-1} \dot{x}_0 = \begin{bmatrix} 20 & 0 \\ 0 & 30 \end{bmatrix}^{-1} \begin{bmatrix} 0,5 & 0,5 \\ -0,5 & 0,5 \end{bmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$x(t) = \sum_{j=1}^{2} \left(a_j \cos(\omega_{nj} t) + b_j \operatorname{sen}(\omega_{nj} t) \right) \phi_j = \begin{pmatrix} 1 \\ 1 \end{pmatrix} 0,5 x_0 \cos(20t) + \begin{pmatrix} -1 \\ 1 \end{pmatrix} (-0,5) x_0 \cos(30t)$$

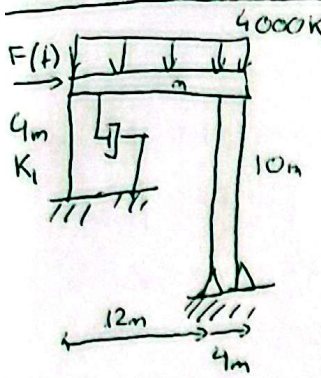
Escaneado con CamScanner

Tema 3: Ejercicio 7

Tema 2: Ejercicio 34

$F(t) = 6000\cos(10t)$ $\quad \xi = 0{,}05 \Rightarrow$ 32576 Nm

$t = 3s \begin{cases} x = 0{,}015\,m \\ \dot{x} = 3\,m/s \end{cases}$



$m = 4000\frac{Kgf}{m} \cdot 16m = 64.000\,Kgf$

$E = 2\cdot10^6\,Kgf/cm^2$

$I_1 = 40000\,cm^4$

$I_2 = I_3 = 16000\,cm^4$

$K_1 = \frac{12EI}{L^3} = \frac{12\cdot2\cdot10^6\,Kgf/cm^2 \cdot 40000\,cm^4}{(400)^3\,cm^3} = 1500\frac{Kgf}{cm}$

$K_2 = \frac{3EI}{L^3} = \frac{3\cdot2\cdot10^6\cdot16000}{(1000)^3} = 96\,Kgf/cm$

$K_{TOT} = 1692\,Kgf/cm$

$\omega_n = \sqrt{\frac{K\left[\frac{Kgf}{cm}\right]}{m[Kgf]}\cdot g\left[\frac{cm}{s^2}\right]} = \sqrt{\frac{1692\cdot981}{64000}} = 5{,}09\,rad/s$

$\omega_a = \omega_n\sqrt{1-\xi^2} = 5{,}08\,rad/s$

$\varphi = \arctan\left(\frac{x_0\cdot\omega_a}{v_0 + \xi\cdot\omega_n\cdot x_0}\right) = 0{,}0254\,rad$

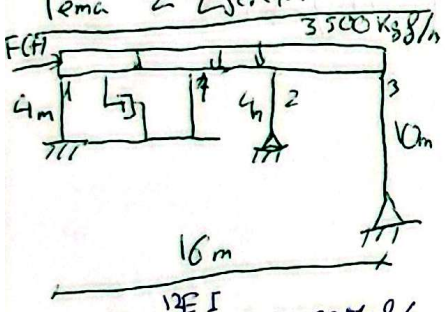
$x_p(t) = \frac{F_0}{\sqrt{(K-m\omega^2)^2 + C^2\omega^2}}\cos\left(\omega t - \arctan\left(\frac{C\omega}{K-m\omega^2}\right)\right)$

$\Rightarrow x_p(t) = 0{,}00126\cos(10t + 0{,}069)$

$x(t=3s) = Be^{-\xi\omega_n t}\cdot\operatorname{sen}(\omega_a t + \varphi) \Rightarrow 0{,}015 = B\cdot0{,}2 \Rightarrow B = 0{,}075\,m$

$x(t) = x_h(t) + x_p(t) = 0{,}075e^{-0{,}2545t}\operatorname{sen}(5{,}08t + 0{,}0254) + 0{,}00126\cos(10t + 0{,}069)\,m$

Tema 2: Ejercicio 35



$E = 2\cdot10^6\,Kgf/cm^2$ $\quad F(t) = 5000\cos(15t)$

$I_1 = 6000\,cm^4$ $\quad \xi = 0{,}12$ $\quad t = 5s \begin{cases} x = 0{,}8\,m \\ \dot{x} = 2\,m/s \end{cases}$

$I_2 = 12000\,cm^4$

$I_3 = 8000\,cm^4$

$m = 3500\cdot16 = 56.000\,Kgf$

$K_1 = \frac{12EI}{L^3} = 2250\,Kgf/cm$ $\quad K_2 = \frac{3EI}{L^3} = 1125\,Kgf/cm$ $\quad K_3 = \frac{3EI}{12} = 48\,Kgf/cm$

$K_{TOT} = 2K_1 + K_2 + K_3 = 5673\,Kgf/cm$

$\omega_n = \sqrt{\frac{K[Kgf/cm]\cdot g[cm/s^2]}{m[Kgf]}} = 9{,}96\,rad/s$

$\omega_a = \omega_n\sqrt{1-\xi^2} = 9{,}89$

$C = \xi\cdot2m\omega_n = 133862{,}4\,N/m$

$\varphi = \arctan\left(\frac{x_0\cdot\omega_a}{v_0 + \xi\cdot\omega_n\cdot x_0}\right) = 0{,}039$

$x(t) = x_h(t) + x_p(t)$

$x_h(t) = Be^{-\xi\omega_n t}\cdot\operatorname{sen}(\omega_a t + \varphi) \Rightarrow x(t=5s) = 0{,}008\,m = B\cdot(-0{,}001779) \Rightarrow B = -4{,}496\,m$

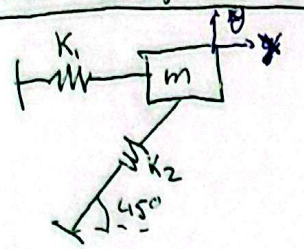
$x_p(t) = \frac{F}{\sqrt{(K-m\omega^2)^2 + C^2\omega^2}}\cos\left(\omega t - \arctan\left(\frac{C\omega}{K-m\omega^2}\right)\right) = 0{,}00068\cos(15t + 0{,}28)$

($F = 5000N$, $\omega = 15\,rad/s$)

$x(t)[m] = -4{,}496e^{-1{,}1952t}\operatorname{sen}(9{,}89t + 0{,}039) + 0{,}00068\cos(15t + 0{,}28)$

Escaneado con CamScanner

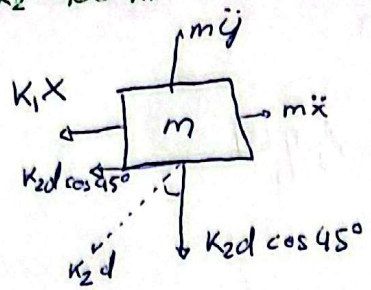
Tema 3: Ejercicio 7



$K_1 = 3000\ N/m$

$K_2 = 4000\ N/m$

$M = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$



$m\ddot{x} = K_1 x + K_2 \frac{x}{2} + K_2 \frac{y}{2}$

$m\ddot{y} = K_2 \frac{y}{2} + \frac{K_2 x}{2}$

$K = \begin{bmatrix} K_1 + \frac{K_2}{2} & K_2/2 \\ K_2/2 & K_2/2 \end{bmatrix} = \begin{bmatrix} 500 & 200 \\ 200 & 200 \end{bmatrix}$

$d = \sqrt{x^2 + y^2}$

$= \left(x\frac{\sqrt{2}}{2} + y\frac{\sqrt{2}}{2}\right)$

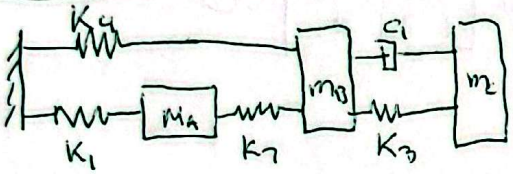
$|K - \omega^2 M| = 0 \Rightarrow \omega \begin{cases} \omega_1 = 10\ rad/s \\ \omega_2 = 25{,}49\ rad/s \end{cases}$

$\phi_1 = \begin{pmatrix} -0{,}5 \\ 1 \end{pmatrix} \quad \phi_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$m_{eff} = \frac{[\phi]^T[M][1]}{[\phi]^T[M][\phi]} = \frac{\begin{pmatrix} 0{,}5 \\ 3 \end{pmatrix}}{\begin{pmatrix} 1{,}25 & 0 \\ 0 & 5 \end{pmatrix}}$

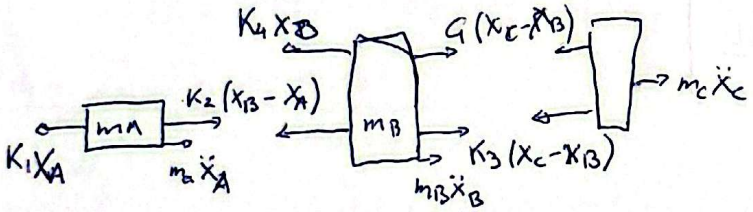
$\begin{cases} m_{eff1} = 0{,}5^2/1{,}25 = 0{,}2\ Kg = 10\% \\ m_{eff2} = 3^2/5 = 1{,}8\ Kg = 90\% \end{cases}$

Tema 4: Ejercicio 2



$[M] = \begin{pmatrix} m_A & 0 & 0 \\ 0 & m_B & 0 \\ 0 & 0 & m_C \end{pmatrix}$

$[K] = \begin{bmatrix} K_1+K_2 & -K_2 & 0 \\ -K_2 & K_2+K_3+K_4 & -K_3 \\ 0 & -K_3 & K_3 \end{bmatrix}$



$[C] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & c_1 & -c_1 \\ 0 & -c_1 & c_1 \end{bmatrix}$

$|K - \lambda M| = 0 \Rightarrow \begin{vmatrix} K_1+K_2-\lambda m_A & -K_2 & 0 \\ -K_2 & K_2+K_3+K_4-\lambda m_B & -K_3 \\ 0 & -K_3 & K_3-\lambda m_C \end{vmatrix} = 0$

$\Rightarrow$ ....

$[K - \omega_n^2 M][\phi_n] = 0 \Rightarrow \phi_n \dots$

Superposición modal

$[M^*] = [\phi]^T[M][\phi] \quad [C^*] = [\phi]^T[C][\phi] \quad [K^*] = [\phi]^T[K][\phi]$

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1 —

Tengo 3 problemas de 1 gdl

Calculamos sus desplazamientos (como en el tema 2); y sus modos propios
$\eta_j(t)$ ϕ_j

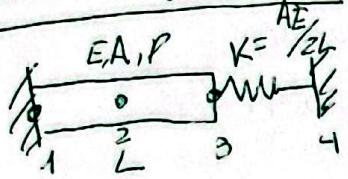
$\eta_1(t) = \dots$

$\eta_2(t) = \dots$

$\eta_3(t) = \dots$

$$x(t) = \sum_{j=1}^{3} \phi_j \eta_j(t)$$

Tema 5: Ejercicio 5



$$[K]_{12} = \frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{matrix} x_1 \\ x_2 \end{matrix} \qquad [K]_{23} = \begin{matrix} x_2 & x_3 \\ \end{matrix}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{matrix} x_2 \\ x_3 \end{matrix}\frac{EA}{L}$$

$$[K]_{mue} = \begin{matrix} x_3 & x_4 \\ \end{matrix}\begin{bmatrix} 1/2 & 0 \\ 0 & 0 \end{bmatrix}\begin{matrix} x_3 \\ x_4 \end{matrix} \cdot \frac{EA}{2L}$$

$$[M]_c = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 2 \end{bmatrix}\frac{\rho AL}{6}$$

$$K_{TOT} = \begin{matrix} & x_1 & x_2 & x_3 \\ x_1 \\ x_2 \\ x_3 \end{matrix}\begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 3/2 \end{bmatrix}\frac{EA}{L}$$

empotr.

$$K_{correg} = \begin{bmatrix} 2 & -1 \\ -1 & 3/2 \end{bmatrix}\frac{EA}{L}$$

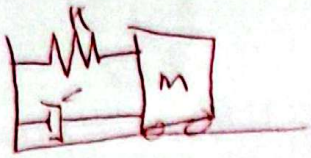
$$\left| \begin{bmatrix} 2 & -1 \\ -1 & 3/2 \end{bmatrix}\frac{EA}{L} - \omega^2 \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}\frac{\rho AL}{6} \right| = 0$$

$$\begin{vmatrix} 2EA/L - \frac{4\omega^2 \rho AL}{6} & -\frac{EA}{L} - \omega^2 \frac{\rho AL}{6} \\ -\frac{EA}{L} - \omega^2 \frac{\rho AL}{6} & -\frac{3EA}{2L} - 2\omega^2 \frac{\rho AL}{6} \end{vmatrix}$$

P_c

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Tema 2: Ejercicio 15



$m = 20\,kg \quad K = 800\,N/m \quad c = 1$

$t = 2\,s \begin{cases} x = 3\,mm \\ V = 2\,m/s \end{cases}$

$\omega_n = \sqrt{K/m} = 6{,}32\ rad/s$

$\omega_a = \omega_n \sqrt{1-\zeta} = 6{,}307$

$\zeta = \frac{c}{2 \cdot m \omega_n} = \frac{1}{2 \cdot 20 \cdot 6{,}32} = 0{,}00395 < 1$ Subamort.

$\varphi = \arctan\left(\frac{x_0 \omega_a}{v_0 + \zeta \omega_n x_0}\right) = 0{,}0158\ rad$

$x = B e^{-\zeta \omega_n t} \operatorname{sen}(\omega_a t + \varphi) \Rightarrow 0{,}005 = B e^{-0{,}05} \operatorname{sen}(12{,}63) \Rightarrow B = 0{,}0827$

Tema 1: Ejercicio 11

$m = 2\,kg \quad T = 4\,s \quad \varphi = 0{,}8\,rad \quad t = 2\,s \quad V = -3{,}3\,m/s$ Extremo sup $\Rightarrow x = A/2$

a) ¿x(t)?

$\omega_n = 2\pi/4 = 1{,}57\ rad/s = \sqrt{\frac{K}{m}} \Rightarrow K = m \cdot \omega_n^2 = 4{,}93\ N/m$

$\frac{1}{2} m v^2 + \frac{1}{2} K x^2 = \frac{1}{2} K A^2 \Rightarrow \frac{1}{2} \cdot 2 \cdot 3{,}3^2 + \frac{1}{2} 4{,}93 \frac{A^2}{4} = \frac{1}{2} 4{,}93 A^2 \Rightarrow \frac{10{,}89}{1{,}85} = A^2 \Rightarrow$

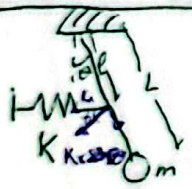
$\Rightarrow A = 2{,}43\ m \Rightarrow x(t) = 2{,}43 \operatorname{sen}(1{,}57 t + 0{,}8)$

b) E y E_K $t = 1{,}5\,s$

$x(t = 1{,}5) = -0{,}0326\ m$

$E_T = \frac{1}{2} K A^2 = 5{,}99\,J \qquad E_K = E_T - E_P = E_T - \frac{1}{2} K x^2 = 5{,}98\,J$

Tema 2: Ejercicio 10



Momentos en O

$J\alpha = \Sigma M_O \Rightarrow m L^2 \ddot{\theta}(t) = -mgL \operatorname{sen}\theta - K x \cos\theta \ell \Rightarrow$

$\Rightarrow m L^2 \ddot{\theta}(t) = -mgL \operatorname{sen}\theta(t) - K\ell \operatorname{sen}\theta \cos\theta \ell \Rightarrow$

$x = \ell \operatorname{sen}\theta$

$\Rightarrow \ddot{\theta}(t) + \theta(t)\left(\frac{mgL + K\ell^2}{mL^2}\right) = 0 \Rightarrow \ddot{\theta}(t) + \theta(t)\left(\frac{g}{L} + \frac{K\ell^2}{mL^2}\right)$

↓ Aprox. ángs pq $\operatorname{sen}\theta \approx \theta$, $\cos\theta = 1$

$\omega_n = \sqrt{\frac{g}{L} + \frac{K\ell^2}{mL^2}}$

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Tema 1: Ejercicio 6

$m = 350\,Kg \quad A = 14\,cm \quad E_m = 15\,J$

a) K? ω_n?

$$E = K + U = \frac{1}{2}mv^2 + \frac{1}{2}Kx^2 = \frac{1}{2}KA^2 \Rightarrow K = \frac{2E}{A^2} = \frac{2 \cdot 15}{0,14^2 m} = 1530,6\ N/m$$

$$\omega_n = \sqrt{\frac{K_{eq}}{m}} = \sqrt{\frac{1530,6}{350}} = 2,09\ rad/s$$

$$f = \omega/2\pi = 0,33\,Hz$$

b) $E_K \quad x = 4\,cm$

$$E = E_K + \frac{1}{2}Kx^2 = 15 = E_K + \frac{1}{2}1530,6 \cdot 0,04^2 \Rightarrow E_K = 13,8\,J$$

Tema 1: Ejercicio 13

Péndulo $A = 0,5m \quad L = 3m \quad \varphi = 0 \quad$ ¿v(t)? ¿T?

$$T = 2\pi\sqrt{L/g} = 2\pi\sqrt{\frac{3}{9,81}} = 3,47\,s$$

$$\omega = 2\pi/T = 1,81\,rad/s$$

$$x(t) = A\,sen(\omega_n t + \varphi) \Rightarrow \dot{x}(t) = A\omega_n \cos(\omega_n t) = 0,41\cos(1,81t)$$

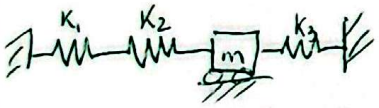
Tema 1: Ejercicio 9

$m_1 = 1000\,Kg$ y $K_1 \quad T = 1s \quad A = 5cm \rightarrow \omega_1 = 2\pi/T = 2\pi\ rad/s = \sqrt{\frac{K_1}{m_1}} \Rightarrow K_1 = 39478\ N/m$

¿m_2? $\quad K_2 = K_1 \quad A_2 = A_1$

$$\omega_2 = 2\omega_1 = \sqrt{\frac{K_1}{m_2}} = 4\pi = \sqrt{\frac{39478}{m_2}} \Rightarrow m_2 = 39478/16\pi^2 = 250\,Kg$$

Tema 2 Ejercicio 5:



$m = 10\,Kg \quad K_1 = 100\,N/m \quad K_2 = 200\,N/m \quad K_3 = 300\,N/m$

$t = 2s \rightarrow x = 20\,mm, \ \dot{x} = 8\,m/s$

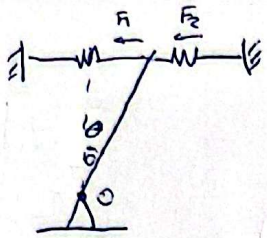
$$K_{eq} = K_3 + \frac{K_1 K_2}{K_1 + K_2} = 394,74\ N/m \quad \omega_n = \sqrt{\frac{K_{eq}}{m}} = 6,28\ rad/s$$

$$\left.\begin{aligned} x &= A\cos(6,28t) + B\,sen(6,28t) \\ \dot{x} &= -A \cdot 6,28\,sen(6,28t) + B\cos(6,28t) \end{aligned}\right\} \Rightarrow \begin{aligned} 0,02 &= A \cdot 1 + B \cdot 0 \Rightarrow A = 0,02 \\ 2 &= -A \cdot 6,28 \cdot 0 + B \cdot 6,28 \Rightarrow B = 0,32 \end{aligned}$$

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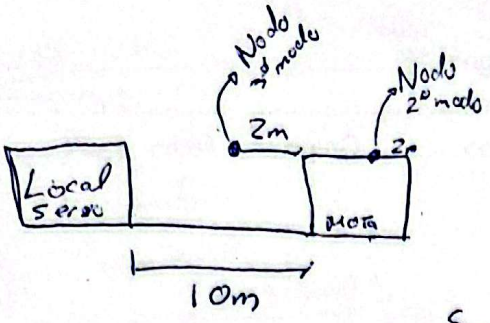
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Alumno ..... n.º.....
Curso ..... Grupo..... Fecha .....



$\Sigma M_O = 0 \Rightarrow I\ddot{\theta}(t) + 2Kx$

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Para el 2º modo → $\frac{720}{14} \approx 52\,KN$

Para el 3er modo → $\frac{720}{8} = 90\,KN$

Se escogería el máximo

Ejercicio 3 (Motores)

6 cil ⇒ Momentos en H

$$623\,KNm / 2m = F_{MH} = 311,5\,KN \approx 315\,KN$$

(2m ↓ separación)

Ejercicio 4 (Motores)

8 cil ⇒ Momentos en X

$L_m = 10,4m \quad L = 2m \quad H_m = 4m \quad M_K = 525\,KNm$

$$F_{Mx} = \frac{525 \cdot 2}{10,4 \cdot 4} = 25,24\,KN \approx 30KN$$

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DE INGENIEROS NAVALES

Asignatura .....
Alumno ..... n.º .....
Curso ..... Grupo..... Fecha .....

Ejercicio 1 (Motores)

$$\tau_{max} = \frac{M_T \cdot R}{\frac{1}{2}\pi R^4} \Rightarrow R = \sqrt[3]{\frac{2M_T}{\tau_{max}\pi}} = \sqrt[3]{\frac{2 \cdot 1,29}{400 \cdot \pi}} = 0,127\,m \Rightarrow D = 254\,mm$$

$$M_T = \frac{P}{N} = \frac{13,5\ MN/s}{\frac{100\pi}{3}\ rad/s} = 1,29\ MNm$$

$$N = 100\ r/m \cdot \frac{2\pi\ rad}{60\ s} = \frac{10\pi}{3}\ rad/s$$

régimen 90% → $P = 15000KW \cdot 0,9 = 13500\ KW = 13,5\ MN/s$

Suponemos $\lambda = 1 \Rightarrow$ Régimen continuo $C_K = 1$ $C_\lambda = 1,38$

$$\tau_1 = \frac{400+160}{18} \cdot 1 \cdot (0,35 + 0,93 \cdot 0,254^{-0,2}) \cdot 1,38 = 67,54\ MPa$$ Régimen Continuo

$$\tau_2 = 1,7 \cdot \tau_1 \cdot C_K^{-0,5} = 114,8\ MPa$$ Régimen transitorio

Rigidez $R = \frac{G\pi d^4}{32L}$

$R_{eq} = R_{ec} \rho^2$ $\rho = \frac{RPM_{hélice}}{RPM_{motor}}$

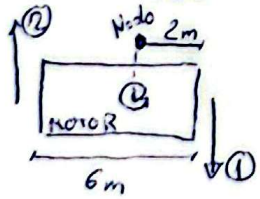
Con reductora

$$I_{red} = I_{ejeintermedio} + I_{piñones} + I^2_{ruedapiñones}$$

$$I_{hélice_{eq}} = (I_h + I_{eje_{ec}} + I_{agua\,arrast})\rho^2$$

Ejercicio 2 (Motores)

$$PRU = \frac{720 \cdot 10^3}{6000} = 120 \rightarrow$$ Recomendable, se usará compensador



① → $\frac{720}{2} = 360\ KNm$

② → $\frac{720}{4} = 180\ KNm$

Escaneado con CamScanner

E. ... 5

Excitaciones a Bordo

Las f_{rec}'s de los diferentes elementos no pueden coincidir (conservador) → ~20% de diferencia entre ellas

$\omega_{superestruc}$

$\omega_{buque,vertical}$

$\omega_{pala} = n \cdot \frac{z \cdot N(rpm)}{60}$ [Hz]

ω_{motor} → $2T = n^{\circ}cil \cdot \frac{N(rpm)}{60}$ [Hz]

$4T = \frac{n^{\circ}cil}{2} \cdot \frac{N(rpm)}{60}$ [Hz]

- Vibraciones del buque

Se estudian: Horizontal, vertical (+ importante) y torsional → Son las frecuencias más bajas

Ejercicio 3

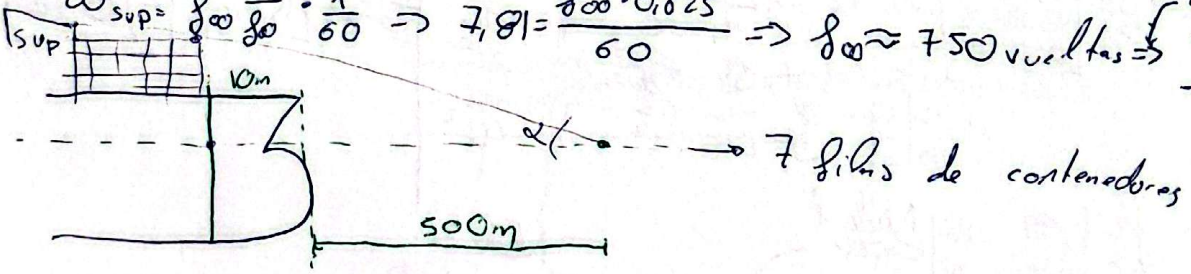
$\left.\begin{array}{l} \omega_{motor} + \omega_{hélic} = 21,86\ Hz \\ \omega_m - \omega_{hél} = 8,2 Hz \end{array}\right\} \Rightarrow \begin{cases} \omega_{motor} = 15,03 Hz \\ \omega_{hélic} = 6,83 Hz \end{cases}$

$rpm_{motor} = rpm_{hélice} \Rightarrow \omega_{motor} = 15,03 = 11 \cdot \frac{rpm}{60} \Rightarrow N = 82 rpm$

$\omega_{sup} = 0,52\, \omega_{motor} = 7,82 Hz$

Sup estruct tipo A: $\frac{f_e}{f_0} = 0,625$

$\omega_{sup} = f_\infty \frac{f_e}{f_0} \cdot \frac{1}{60} \Rightarrow 7,81 = \frac{f_\infty \cdot 0,625}{60} \Rightarrow f_\infty \approx 750$ vueltas $\Rightarrow$ (Tabla) $h_{sup} = 17,5 m$



Escaneado con CamScanner

Sistemas Continuos de Masas

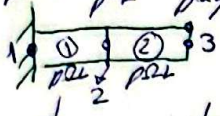
Usaremos la matriz de masa consistente → $[M] = \iiint \rho [N]^T [N] dV$

Funciones de forma que dependen de las condiciones de contorno

Ejercicio 3

Freely supported a carga axial discretizada en 2 elementos para acero y aluminio

$\rho AL/2$ ρAL $\rho AL/2$ Elemento uniaxial con carga axial



$[K_1] = \frac{EA}{L_1}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}$ $[K_2] = \begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}\frac{EA}{L_2}$

Masas concentradas

ENSAMBLAMOS → $[K] = \begin{bmatrix}1 & -1 & 0\\ -1 & 2 & -1\\ 0 & -1 & 1\end{bmatrix}$ elimino: el nodo 1 no se puede desplazar

$[M] = \frac{\rho AL}{2}\begin{bmatrix}0 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 1\end{bmatrix}$

$\det([K] - \lambda[M]) = 0 \Rightarrow \left|\frac{AE}{L}\begin{pmatrix}2 & -1\\ -1 & 1\end{pmatrix} - \lambda\frac{\rho AL}{2}\begin{pmatrix}2 & 0\\ 0 & 1\end{pmatrix}\right| = 0$

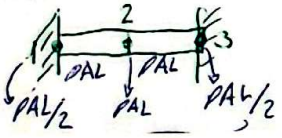
Acero: $\omega_1 = 1{,}241 \cdot 10^3$ rad/s $\omega_2 = 2{,}29 \cdot 10^3$ rad/s

Aluminio $\omega_1 = 1{,}32 \cdot 10^3$ rad/s $\omega_2 = 2{,}45 \cdot 10^3$ rad/s

$K_1 = \begin{bmatrix}K_{11} & K_{12}\\ K_{21} & K_{22}\end{bmatrix}$

$K_2 = \begin{bmatrix}K_{22} & K_{23}\\ K_{32} & K_{33}\end{bmatrix}$

Ejercicio 6



$[K_1] = \frac{EI}{L^3}\begin{bmatrix}12 & 6L & -12 & 6L\\ 6L & 4L^2 & -6L & 2L^2\\ -12 & -6L & 12 & -6L\\ 6L & 2L^2 & -6L & 4L^2\end{bmatrix} = [K_2]$

Acoplada

	v_1	θ_1	v_2	θ_2	v_3	θ_3
v_1	12	6L	-12	6L	0	0
θ_1	6L	$4L^2$	-6L	$2L^2$	0	0
v_2	-12	6L	24	0	-12	6L
θ_2	6L	$2L^2$	0	$8L^2$	-6L	$2L^2$
v_3	0	0	-12	-6L	12	-6L
θ_3	0	0	6L	$2L^2$	-6L	$4L^2$

$\frac{EI}{L^3} \Rightarrow [K_{correg}] = \begin{bmatrix}24 & 0\\ 0 & 8L^2\end{bmatrix}\frac{EI}{L^3}$

Nodo 1 empotrado

Nodo 3 empotrado

$[M_2] = [M_1] = \begin{bmatrix}156 & 22L & 54 & -13L\\ 22L & 4L^2 & 13L & -3L^2\\ 54 & 13L & 156 & -22L\\ -13L & -3L^2 & -22L & 4L^2\end{bmatrix}\frac{\rho AL}{420}$

$[M_{acopl}] = \begin{bmatrix}156 & 22L & 54 & -13L & 0 & 0\\ 22L & 4L^2 & 13L & -3L^2 & 0 & 0\\ 54 & 13L & 312 & 0 & 54 & -13L\\ -13L & -3L^2 & 0 & 8L^2 & 13L & -3L^2\\ 0 & 0 & 54 & 13L & 156 & -22L\\ 0 & 0 & -13L & -3L^2 & -22L & 4L^2\end{bmatrix}$

$\Rightarrow [M_{correg}] = \begin{bmatrix}312 & 0\\ 0 & 8L^2\end{bmatrix}\frac{\rho AL}{420}$

$([K] - \omega^2[M])\{x\} = 0$

Escaneado con CamScanner

Ejercicio 5

$$P_z \leq \frac{8000\ N/m^2}{2,5} \leq 3200\ N/m^2$$

$$P_z = \sqrt{P_o^2 + P_c^2} \Rightarrow P_c = \sqrt{P_z^2 - P_o^2} = 3025,05\ N/m^2$$

$$P_o = \frac{(100 \cdot ,4)^2}{70 \cdot 4^{1,5}} \left(\frac{2,02}{1,1/2}\right) = 1049\ N/m^2$$

$d = 0,9 + 0,1 \cdot 2 = 1,1$

$K_o = 1,8 + 0,4 \cdot \left(1,1/2\right) = 2,02$

$$P_c = \frac{(N D_p)^2 \, V \cdot (1-w)}{160\sqrt{h_a + 10,4}} \Rightarrow h_a = \left(\frac{(100 \cdot 4)^2 \cdot 15 \cdot 0,514\,(1-0,3)}{160 \cdot 3023,05}\right)^2 - 10,4 = 7,82m$$

(K_c / d/R)

$d/R < 1 \Rightarrow K_c = 1,7 - 0,7(d/R)$

$T = h_a + \text{Altura eje} = 10,82m$

$\Delta = BLT \cdot C_B \cdot \rho = 56 \cdot 375 \cdot 0,62 \cdot 1,025 \cdot 10,82 = 144398,3\ t$

$\Delta = PM \cdot AR \Rightarrow PM = 124398\ t = 121364\ m^2$

¿ en este caso el buque

buque $\rho = 1,025\ t/m^3$

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Ejercicio 1 (Hélices)

$12\ Kn \cdot \frac{0,514\ m/s}{1\ Kn} = 6,168\ m/s$

$20\,000\ KW \cdot \frac{1\ CV}{0,7457\ KW} = 26820\ CV$

Hay que aplicarle el régimen de trabajo del motor: ~90%

$DHP = \frac{BHP}{1,026} \cdot \eta_m \cdot \eta_r = \frac{26820}{1,026} \cdot 0,95 \cdot 1 = 24834\ CV$

$T = \frac{DHP \cdot 75 \cdot \eta_o}{v(1-\omega_T)} = \frac{24834 \cdot 75 \cdot 0,7}{6,168\,(1-0,4)} = 352294\ Kg$

$Q = \frac{T \cdot v(1-\omega_T)}{2\pi N \eta_o} = \frac{352294 \cdot 6,168\,(1-0,4)}{2\pi \cdot 6000 \cdot 0,7} = 19,4\ Kg\,m$ → 177503 Kg m

100 rpm = 6000 rps → 100/60 rps = 1,67 rps

1,67

Ejercicio 2 (Hélices)

Como ~~la~~ opción 1 el nº cilindros es múltiplo del nº pala cuadadocantado
Se coge la opción 2 ya que las frecuencias de motor y hélice estarán más alejadas

Ejercicio 3 (Hélices)

El portacontenedores al tener formas en V genera más velocidad y por tanto más vibraciones (un menor ω_T)

Porta 1 hélice $\omega_T = 0,5 \cdot C_B - 0,05 = 0,5 \cdot 0,6 - 0,05 = 0,25$

Petrol 2 hélices $\omega_T = 0,55 C_B - 0,2 = 0,24$ → Peor

Ejercicio 4

4 palas:

$\omega_{pala} = \frac{4 \cdot N(rpm)}{60} = 7,67\ Hz$

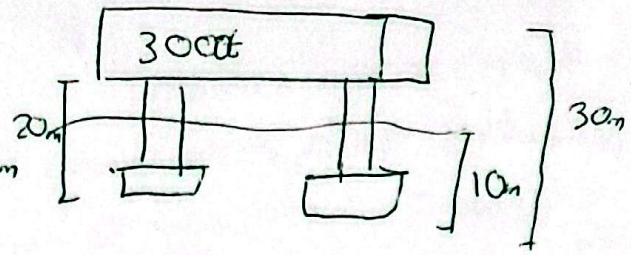
5 palas

$\omega_{pala} = \frac{5 \cdot N}{60} = 9,58\ Hz$

Amplitud armónico:
- $\bar{X}$ → $A = K2 = 4, 8, 12 \ldots$
- $\bar{Y}, \bar{Z}$ → $A = K(Z \pm 1) = 3, 5, 7, 9 \ldots$

Escaneado con CamScanner

Ejercicio 11



¿GM > 0?

$P_{Hull} = 5000t$

$KG_{Hull} = 10m$ $\quad KB = 5m$

$\nabla_{10m} = 10.000m^3$

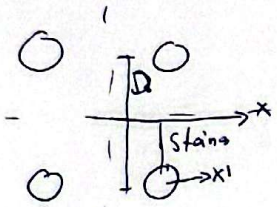
$KG_{Lastre} = 2m$

$I_{44} = 1,654 \cdot 10^{10} kg\,m^2$

$GM = KB + BM - KG$

$BM = \frac{I}{\nabla}$

$I_{x1} = \frac{\pi R^4}{4}$



$\Rightarrow I_x = 8\pi + 9\pi D^2$

$\Delta = P_{Hull} + P_{TOP} + P_{Lastre} \Rightarrow$

$\Rightarrow P_{Lastre} = \nabla\rho - P_{Hull} - P_{TOP} \Rightarrow$

$\Rightarrow P_{Lastre} = \left(1,026 \frac{t}{m^3} \cdot 10000 m^3\right) - 3000 - 5000$

$\Rightarrow 2260t$

	Pesos (t)	KG (m)	P·KG (t·m)
Hull	5000	10	50000
Topside	3000	30	90000
Lastre	2260	2	4520

$KG = \frac{\Sigma P KG}{\nabla\rho} = 14,09m$

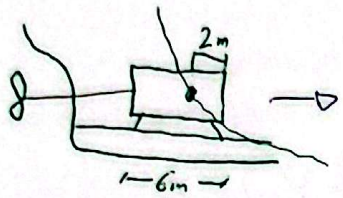
$GM = KB + BM - KG > 0 \Rightarrow 0 = 5 + \frac{8\pi + 9\pi D^2}{10.000} + 14,09 \Rightarrow D \simeq 57m$

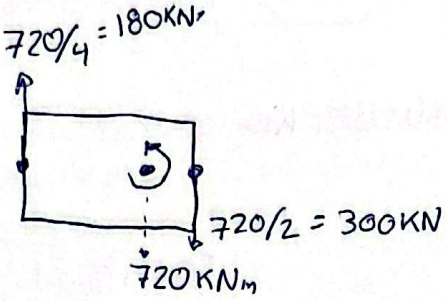
Escaneado con CamScanner

Ejercicio 6

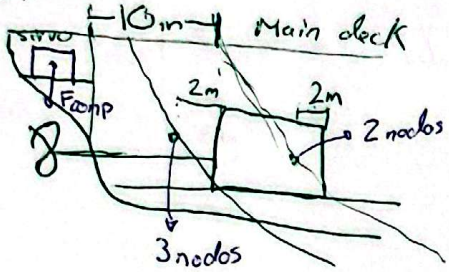
$PRU = \frac{720}{6000} = 120 \Rightarrow$ Recomendable → Se pone

a) ¿Interior?





b) ¿Exterior?



2 nodos: $F_{comp} \simeq 52$ KN

3 nodos: $F_{comp} = 90$ KN

} cogemos la mayor

N = 200 rpm (el doble de rpm) → Tabla ⇒ KT200VA

Tarea: Ejercicio 8

Pot necesaria: 2000 KW

a) Elegir un motor con régimen de trabajo de 85%

Pot motor = $2000/0{,}85 = 2353$ KW = $P_{máx}$

Elegimos Yanmar 8EY26W → 4 tiempos Pot. máx = 2360 KW / rpm = 750 → reductora 2,8

11

12,5 Hz

b) ¿$\omega_{hélice}$? ¿4 y 5 palas cuál es mejor?

$rpm_{hélice} = 750/2{,}8 = 267{,}86$ rpm

$\omega_{pala} = n \cdot \frac{z \cdot N}{60} = 1 \cdot \frac{4 \cdot 267{,}86}{60} = 17{,}86$ Hz

$L_o = 1 \cdot \frac{5 \cdot 267{,}86}{60} = 22{,}32$ Hz → Es mejor porque frec. más alta → 20% = 26,8

$\omega_{Viga} = (\beta L)^2 \sqrt{\frac{EI}{mL^4}} = 1{,}875 \sqrt{\frac{EI}{mL^4}} > 26{,}8$ Hz $= 1{,}875 \cdot \sqrt{\frac{121\,540 \cdot 10^6 \text{ Kg/cm}^2}{L^4 \cdot \rho L \cdot 19{,}75 \text{cm}^2}} \Rightarrow$

↓ Empotrado-libre Viga HSS

$\Rightarrow L^{5/2}_{[m]} = \frac{1875}{26{,}8} \sqrt{\frac{119 \cdot 10^{15} \text{ N/m}^2}{7{,}8 \text{ Kg/m}^3 \cdot 0{,}002 \text{m}^2}}$

$\Rightarrow L^{5/2} = 61{,}5 \cdot 10^6 \Rightarrow L =$

Seleccionamos viga HSS20×0,5 → D: 20 inches = 508 mm $I = 1460$ in^4 = 607 697 882 mm^4

t = 0,5 inches = 12,7 m

A = 30,6 in^2 = 19 741,9 mm^2 = 19,75 cm^2

" 0,000608 m^4

" 60 769,78 cm^4

$E = 2 \cdot 10^6$ Kg/cm^2

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a)

$$\omega_{motor} = n^{\circ}cil \cdot \frac{N}{60} = 7 \cdot \frac{105}{60} = 12{,}25\ Hz$$

N(rpm) = 105 rpm
2 tiempos
nº cil

$$\omega_{pala} = n \cdot \frac{2 \cdot N(rpm)}{60} = 1 \cdot \frac{4 \cdot 105}{60} = 7\ Hz$$

N(rpm) = 105 rpm
Z = Estos buques suelen tener del orden de 4-5 palas, ningún número es múltiplo del ~~orden~~ número de cilindros del motor, pero se escoge 4 palas para que las frecuencias estén lo más alejadas.

Para la $\omega_{superestructura}$ queremos que esté un 20% alejada de las otras dos frecuencias

$$14{,}7\ Hz < \omega_{sup} < 9{,}8\ Hz$$

$$8{,}4 < \omega_{sup} < 5{,}6\ Hz$$

Puede estar entre 8,4 y 9,8 Hz pero en el caso de que finalmente se decida una hélice de 5 palas podría haber problemas. Por lo que supondremos de $\omega_{sup} > 14{,}7$ Hz. Porque si es más pequeña puede dar problemas

$$\omega_{sup} > 14{,}7\ Hz$$

$$\omega_{sup} = f_{\infty} \cdot \frac{f_e}{f_{\infty}}\ [rpm] \Rightarrow 14{,}7 \cdot 60 = f_{\infty} \cdot 0{,}625 \Rightarrow f_{\infty} = 1411{,}2$$

↳ ¡Queda fuera de la línea en la gráfica por lo que seguro será < 10m!

↳ Del plano de envolvente vemos que es una superestructura Tipo C

Si comprobamos con un valor entre ambas frecuencias por ejemplo 9Hz

$f_{\infty} = 864 \Rightarrow h_{meters} = 15_m$

Con esta solución, no se podría poner una ~~pala~~ hélice de 5 palas!

Escaneado con CamScanner

ANSYS

- Arrastrar / Doble click en el módulo [Static Structural]
- Click dcho en Geometry (para dibujar) → Design Modeler
- SKetching
- Poner 2 nodos y LinKarks
  - Modeling → Create → Point → Definition → Manual input → Generate
  - Unirlo → Concept → Lines from points → Clikar los ptos con Ctrl → Generate
  - Sección → Concept → Cross section → Circular o elegir otra
  - Volver a la línea → Cross Section y elegir la correspondiente

Cerramos y Geometry con tick verde

Cliko dos. veces en Model (para calcular)

- Display → Thick Shells & Beam + Cross section
- Ver izqda → Mesh, click dcho → Generate Mesh
- cc → Click dcho en Static Structural → Insert (radio contorno) → Fixed support = Empotramiento / Displacement = Apoyos (△)
  - → Ir a la barra de arriba y clicar sobre vértice → Clicar en vértice y aplicar (⇆)
- Fuerza → Click dcho en Static Struct → Insert → Force
  - Define By → Components (se escribe) → Clicamos en Vértice

Si pinchamos Static Structural nos muestra lo que hay

- Clik dcho Solution → Insert → Deformation, Beam Results,
- Solution (arriba del todo) → Solve

Dinámica: Modal → Arrastrarlo hasta Geometry

- Cliko sobre Model
  - En Modal → Cond Contorno
  - Solution → Insert → Deformatio
  - Analysis settings → Max Modes to find (seleccionar cuantos)
  - Solve → Clikar Sobi
  - Total Deformation → Create results at all sets → Volver a Solve

Escaneado con CamScanner

Solution information (textaco de como trabaja)

- Frecuencias
- Factores de participación modal

Seleccionar materiales

- Egineering Data → Engineering Data Sources → General Materials → ⊕ Añadi.

PLANCHAS

Hacer plancha

- Geometry → Design Modeler → Crear Punto → Concept → line from points
- Concept → Surface from edges → Borrar líng
- Modal → Insert → Cond contorno

Escaneado con CamScanner